



Modeling microstructural evolution in irradiated materials with cluster dynamics methods: A review [☆]

Aaron A. Kohnert ^{a,*}, Brian D. Wirth ^{b,c}, Laurent Capolungo ^a

^a Los Alamos National Laboratory, Los Alamos, NM 87545, United States

^b University of Tennessee, Knoxville, TN 37996, United States

^c Oak Ridge National Laboratory, Oak Ridge, TN 37830, United States

ARTICLE INFO

Article history:

Received 31 October 2017

Received in revised form 22 February 2018

Accepted 23 February 2018

Available online 29 March 2018

Keywords:

Cluster dynamics

Radiation damage

ABSTRACT

As both the design of material systems for nuclear energy applications and the number of reactor designs being proposed increase the need for material models predicting the evolution of microstructures as a function of chemistry, texture, grain size, precipitate content, etc., and irradiation conditions is steadily increasing. This manuscript aims at presenting a review of the different cluster dynamics modeling schemes that have emerged worldwide over the past decade(s). Further, the manuscript also critically discusses limitations in existing approaches and identifies potential routes for future developments.

© 2018 Elsevier B.V. All rights reserved.

1. Introduction

Materials subject to irradiation often undergo instantaneous microstructural changes which over the long term can induce changes to physical and mechanical properties, potentially compromising reactor and component performance on an engineering scale. Void induced swelling is typically one of the most visible effects of irradiation as is a significant downward shift in the ductile to brittle transition temperature. Both the increasing complexity in the material design and the pragmatic limitations in gathering experimental data representative of neutron irradiation motivate the need for advanced models predicting microstructure evolution over years of sustained irradiation as a function of both irradiation conditions (i.e. dose, dose rate, temperature), material chemistry and initial microstructure (e.g. grain size).

Evolution on the microstructural scale is a consequence of the transport of point defects generated in energetic recoil events to various sinks within the microstructure, the segregation and aggregation of solutes as induced or accelerated by the supersaturations of such defects, and the accumulation and growth of sessile defect clusters. The exact response of a material depends on the conditions of exposure, such as temperature and radiation intensity, in addition to the properties of the material itself such as point defect energetics, chemistry and crystal structure, and the pre-existing microstructure. Theoretical descriptions of radiation damage pro-

cesses are complex, and must span the relevant time and length scales inherent in the problem from point defect generation and migration in femtosecond events on the atomic scale to property changes which occur over the course of years or decades throughout nuclear reactor components on the engineering scale. The former is best handled through atomistic models, with density functional theory and molecular dynamics providing a wealth of information about energetic recoil cascades, point defect formation and migration energies, and other important aspects of the atomic level aspects of point defect behavior.

Changes in the physical and mechanical properties which might compromise the performance and reliability of structural materials result from the collective action of these defects, and understanding such processes requires a theoretical framework with a broader view than the atomic level is generally capable of providing. Evolution of the microstructure is governed by the accumulation, transport, and elimination of defects, as well as their mutual interactions, and interactions with the existing microstructure. Addressing the effects which occur over longer time scales (microns) and length scales (years) requires a model which can describe the density of defects which a particular irradiation condition will produce, account for the various interactions between them and the microstructure, and ultimately predict the consequent changes in the microstructure that can be expected.

The cluster dynamics (CD) modeling framework is one example of this type of model. Built upon mean field rate theory (MFRT), a fundamental modeling approach in radiation damage, CD has a wide variety of applications in modeling microstructural evolution. In this paper, we illustrate the connection between the CD

[☆] This article belongs to the special issue: Radiation Effects.

* Corresponding author.

E-mail address: akohnert@lanl.gov (A.A. Kohnert).

approach and the more traditional MFRT models. We discuss the advantages that the additional complexity of modern CD models offer over their traditional counterparts, identifying each component of the modeling framework and the physical processes the model describes. A comparison of the various techniques implemented to improve the efficiency of CD simulations follows, accompanied by a discussion of the limitations and comparative advantages of each. Finally, we highlight some areas of application for the model, discussing unique aspects of implementation, current progress, and/or outstanding questions in each.

2. Mean field rate theory

Since its development in the early 1970s, mean field rate theory (MFRT) has been one of the most important conceptual tools for understanding radiation response. The key concepts rely on using the point defect balance equations in order to predict the concentrations of self interstitial atoms (SIA) i and vacancies v . Such can be expressed as follows:

$$\frac{dC_i}{dt} = g_i - k_{i,v}C_iC_v - D_i k_i^2 C_i \quad (1)$$

$$\frac{dC_v}{dt} = g_v - k_{i,v}C_iC_v - D_v k_v^2 C_v \quad (2)$$

where the first term describes defect production during recoil events, the second mutual recombination events which eliminate a defect of both types, and the third the loss of defects at sinks in the microstructure. Here, C is the concentration, D the specie diffusion coefficient (distinct from the self-diffusion coefficient of the material), $k_{i,v}$ a recombination rate constant, and k^2 the sink strength for removal of the defect. These methods were introduced primarily to rationalize the void swelling phenomenon [1–8], but have also been applied extensively to other phenomenon such as segregation [9,10] and radiation creep [11–13].

In MFRT, the point defects are only lost through absorption at sinks or recombination events. The total current of a given point defect into sinks is described by Dk^2C , and the determination of the sink strength is key to the operation of the models. Conceptually, the sink strength appears from replacing a complex microstructure with numerous discrete objects that might absorb point defects with an homogenized lossy medium, which absorbs defects uniformly. The proper magnitude of the sink strength can be found by demanding that the current of point defects computed in the lossy medium matches the current that would be found in the real system, a process which will be discussed in detail later. Prominent examples of sinks include grain boundaries, cavities, dislocations, and precipitates. Each of these classes of sink is typically assigned a partial sink strength that may have different magnitudes for different defects and the total sink strength is then written as the linear superposition of the partial sink strengths, for instance

$$k_v^2 = \sum_j k_{v,j}^2 \quad (3)$$

for the vacancies. The sink strength itself has dimensions of inverse length squared with little in the way of intuitive physical meaning, but the quantities $(Dk^2)^{-1}$ and $1/k$ can be interpreted as the mean lifetime and root mean square displacement between production and absorption at a sink.

When the concentrations of the point defects in the mean field are known, the current of these defects into various sinks can be calculated. For a given sink of type L , the number of vacancies accumulated at that sink can be written by subtracting the inbound SIA current from the vacancy current

$$\frac{dN_{v,L}}{dt} = D_v k_{v,L}^2 C_v - D_i k_{i,L}^2 C_i - (D_v k_{v,L}^2 C_v^0 - D_i k_{i,L}^2 C_i^0) \quad (4)$$

where the last term describes the thermal emission of defects from the sink, with the contribution from interstitials usually negligible. The vacancy and SIA densities can be found from the defect balance equations at steady state. Limiting cases such as the sink dominant condition $Dk^2C \gg k_{i,v}C_iC_v$ can be used to simplify the system, in which case at steady state $g = Dk^2C$. Under such circumstances, one might write

$$\frac{dN_{v,L}}{dt} = g \left(\frac{k_{v,L}^2}{k_v^2} \frac{k_{i,L}^2}{k_i^2} \right) - D_v k_{v,L}^2 C_v^0 \quad (5)$$

if SIA emission can be neglected. In this way, one can predict how a given set of sinks respond to continued radiation exposure, with the method applied perhaps most successfully to the phenomenon of void swelling in stainless steels [14,15].

With the advent of molecular dynamics (MD) simulations of energetic recoil events, it became clear that a significant fraction of defects produced in neutron or heavy ion relevant damage conditions were not monomer SIA and vacancy defects, but rather clusters of defects [16–21]. Prior rate theory considerations of in-cascade clustering had argued that this type of in-cascade clustering would lead only to a decrease in the effective rate of damage production, premised on the notion that any such clusters would be sessile and serve as recombination sites [22]. Rather, MD simulations indicated that many of the clusters produced in this fashion were mobile, and furthermore, that the mobility of the interstitial clusters may be restricted to one dimensional trajectories [23–25]. Concurrently, a more complex vision of earlier rate theory models emerged based on the concept of “production bias” which included separate balance equations for the sessile and one dimensionally migrating defect species [26–31], but treated all the members of each class of defect as identical. More recently, attempts to further refine and more accurately model the rate theory treatment of point defect production, mobility, clustering, and partition between sinks have been carried out within the CD framework.

3. Cluster dynamics

The cluster dynamics method is an expansion of the MFRT models to include clusters of point defects in addition to the SIA and vacancy monomers. In addition to treating these additional species, most modern cluster dynamics simulations consider spatial dependence. The addition of these species alters the fundamental aspects of the point defect balance equations very little, such that for a cluster n ,

$$\frac{dC_n}{dt} = -\nabla \cdot J_n + g_n + R_n(\vec{C}) - D_n k_n^2 C_n \quad (6)$$

with J_n the flux of species n . The recombination term from standard rate theory is replaced with a more general reaction term, which in principle can include interactions with any other cluster in the system. For a given species n ,

$$R_n(\vec{C}) = \sum_{i+j \rightarrow n} k_{ij}^+ C_i C_j - \sum_{\substack{i+n \rightarrow m \\ n+i \rightarrow m}} k_{i,n}^+ C_i C_n + \sum_{\substack{m-i \rightarrow n \\ m-n \rightarrow j}} k_{m,i}^- C_m - \sum_{n-i \rightarrow j} k_{n,i}^- C_n \quad (7)$$

where the first term describes the formation of n through the combination of two reactant clusters, the second the loss of n through the combination of n with other clusters, the third the formation of n through the dissociation of a parent cluster, and fourth the loss of n by the dissociation of other clusters from it. The notation on the

summations of the form $a + / - b \rightarrow c$ indicates the set of all possible (mass conservative) reactions (+) or dissociations (–). Accordingly, $k_{a,b}^+$ is the reaction rate coefficient between a and b , (the recombination rate coefficient for the case i, v) and $k_{a,b}^-$ is the dissociation rate constant for the emission of b from a . The alternate symbols β and α are often used for reaction and dissociation rate coefficients, respectively. Generally, many clusters are capable of reaction and the relevant summations can produce significant coupling between the equations for various clusters. By contrast, dissociation events are usually considered to be restricted to the emission of the monomer species such as single interstitials and vacancies, though in principle more complex events are possible (such as loop punching, the emission of larger interstitial clusters from overpressurized gas bubbles). The extent of reaction network density varies, with simpler models only accounting for SIA and vacancy absorption akin to standard MFRT while others consider interstitial clusters mobile and capable of interaction up to large sizes.

In cluster dynamics, the evolution of certain classes of sinks is handled inherently by the growth of clusters to larger sizes with any change in sink strength reflected in the rate coefficients for the different clusters as will be described in Section 3.2. For example, a cluster dynamics model would include distinct balance equations for the density of vacancy clusters of two, of three, or four, and so on up to thousands, millions, or even billions of constituents rather than a single equation governing the average void size. This provides entire size distributions of sinks, represented as a set of concentrations for clusters of each discrete size. The void evolution is then described as a transfer of mass, from smaller clusters to larger ones by absorbing vacancies, and from larger clusters to smaller ones by absorbing SIAs. Similarly, dislocation loop formation and growth manifests as the growth of the interstitial cluster distribution, and void swelling as the growth of the vacancy cluster distribution. Consequently, nucleation and growth of such sink classes are both described naturally in these models, with no need for imposed feature densities, external nucleation models, or other such assumptions. Other types of sinks, such as network dislocations and grain boundaries, remain external to the cluster distributions and are either included in the sink strength term or treated explicitly in a spatially resolved simulation (as will be discussed later).

The other major advantage of cluster dynamics over its simpler rate theory predecessors is the ability to include cascade damage directly. Molecular dynamics simulations of energetic recoil events have indicated that for all but the lowest knock-on energies much of the damage is produced already in the form of extended clusters, in some cases with sizes extending to several dozen members. Depending on the material and its crystal structure, these clusters may be in the form of small cavities, stacking fault tetrahedra, dislocation loops of either a sessile or glissile character, and perhaps even more exotic lattice defects. Depending on the binding and migration energetics of these clusters, they may behave similarly to Frenkel pair analogues, may serve as immobile sink nuclei and/or recombination centers, or if in the form of a glissile dislocation loop may glide away from the cascade along one-dimensional trajectories. Early attempts to include such phenomenon in the traditional rate theory approach, collectively referred to as “production bias” models, indicated that considering this in-cascade clustering can have profound effects on the point defect partition and ultimately, predictions of microstructural evolution. In the cluster dynamics framework, in-cascade clustering is handled quite naturally, simply by supplying source terms g for as many clusters as necessary with the appropriate size distributions. Once again, cluster dynamics can handle all the subsequent defect

behavior naturally, and for metals where the primary damage state is well characterized, no *a priori* assumptions about which clusters can dissociate to release free diffusing defects need to be made. Where the energetics of defect diffusion and dissociation processes are sufficiently well known, the reaction network of Eq. (7) resolves all such events.

3.1. Spatial resolution

A key development in the application of MFRT based models has been the creation of spatially resolved models. The motivations for these have varied, with efforts to address polycrystalline environments [32], the thin film irradiations performed in *in situ* irradiation experiments [33–35], and the near surface gas implantations relevant to plasma surface interactions [36–38]. Due to the difficulty of posing a weak form of the defect balance equations with reaction terms, such simulations are performed using finite difference discretizations. A one dimensional discretization is sufficient to capture the changes in behavior that occur near planar boundaries and surfaces, and have shown the denuded zones typically observed near such sinks. Two and three dimensional discretization are becoming more common, allowing high fidelity descriptions of defect transport in environments with nanoscale features or sensitivities, and even allowing an analysis of morphology changes as a consequence of exposure. Presently, it is common to include planar sinks such as grain boundaries and free surfaces explicitly in spatially resolved models. In the context of hierarchical scale transitioning, spatially resolved CD can be used to develop reduced order models describing the net effect of sinks. For example, in the case of grain boundaries spatially resolved methods established correlations between point and cluster defects behavior at the material interface, and irradiation conditions. Notwithstanding the potential for these approaches to identify the key interface phenomena/processes leading to spatially varying defect profiles – the formation of depletion zones being a prime example of such– these studies reveal the ambiguity associated with describing material interfaces solely through a sink efficiency [39,40].

Spatial resolution is implemented by expanding the divergence term in the Eq. (6). The flux can be expressed generally in terms of the chemical potential of a defect species μ_n as

$$-J_n = \frac{D_n C_n}{k_b T} \nabla \mu_n. \quad (8)$$

The variation in chemical potential with concentration is well described by

$$\mu_n = \mu_n^0 + k_b T \log(\Omega C_n) \quad (9)$$

with the reference potential μ_n^0 dominated by the local formation energy of the point defect E_n . Accordingly,

$$-J_n = D_n \nabla C_n + \frac{D_n C_n}{k_b T} \nabla E_n \quad (10)$$

where in the radiation effects context, the primary source of gradients in E_n are interactions between the defect and the elastic strain field in the surrounding material, and in standard rate theory are accounted for implicitly rather than explicitly. Other factors can also contribute to the defect energetics, such as variations in the local chemistry of a material. In simulations where any strain inducing sinks are treated as homogenized absorbers rather than discrete features in the mesh and energy gradients vanish, and the transport term reduces to

$$-\nabla \cdot J_n = D_n \nabla^2 C_n + \nabla D_n \cdot \nabla C_n \quad (11)$$

where the diffusivity gradient is usually neglected. The diffusion coefficient is expanded in terms of a prefactor D_0 and migration energy E^m such that

$$D_n = D_{0,n} \exp \frac{-E_n^m}{k_b T} \quad (12)$$

such that the diffusion coefficient will vary spatially if a temperature gradient exists in the material. In cases where temperature gradients are not negligible, spatially resolved cluster models must include the second term in Eq. (11) to account for thermally motivated drift.

Providing spatial resolution essentially allows the assumed concentration fields surrounding sinks in the derivation of sink strengths to be realized directly. As such, the changes in interactions between mobile defects, and the spatial correlation of the nucleation of new sinks with the footprint of existing ones can be taken into account explicitly. These effects cannot be realized with a homogeneous description of the sink as a lossy medium, and in fact any changes to interactions (such as recombination) as a function of position due to changes in the concentration profile are assumed to be negligible in the sink strength derivations [41]. The development of spatially dependent models which account for an increasing proportion of the sinks explicitly rather than implicitly will allow more detailed study into the importance of sink arrangements, shapes, and strain fields on microstructural evolution. We note that the sink strength of explicitly meshed features may still need to be computed in some manner as in some cases the absorption rates at homogenized sinks may depend on many body interactions. Finally, for reactions involving particularly large clusters (with physical dimensions approaching the mesh spacing) the rate coefficients may begin to lose meaning in spatially resolved simulations as the assumptions used to construct them, discussed at length in the next section, begin to break down.

3.2. Reactions and homogenized sinks

The rate of reaction of mobile point defects with the cluster distributions is described by the rate coefficients k^+ in Eq. (7). These coefficients are derived from the assumption of a diffusion limited reaction process, and are thus related to the partial sink strengths of MFRT through the simple equivalence

$$D_i k_{ij}^2 C_i = k_{ij}^+ C_i C_j \quad (13)$$

for diffusing species i and sink species j , where both sides describe the rate of reaction these two species. As such, the reaction rate coefficients required to construct a CD model can be taken directly from, and are fully consistent with, the partial sink strengths of MFRT. Here the total sink strength of a system for absorbing i can be found by a linear summation over the partial sink strengths of all sinks as in Eq. (3). The reaction kinematics of three dimensionally diffusing species with sinks has a long history of study. Approaches vary, but the most widely adopted has been the effective medium approach of Brailsford and Bullough [41]. A more detailed discussion of this method with an example of how to derive sink strengths is provided in Appendix A.

3.2.1. The kinetics of reaction

The partial strength for a spherical sink j (perhaps a particular vacancy cluster or precipitate in the CD context) absorbing a defect i at a capture radius of r_{ij} is given by,

$$k_{ij}^2 = 4\pi r_{ij} C_j (1 + k_i^2 r_{ij}^2) \quad (14)$$

which differs from the classical expression of Smoluchowski [42] only by the second order term $k_i^2 r_{ij}^2$. In many cases, this correction

is negligible, though it becomes significant for larger sinks when the total sink strength is high, as illustrated in Fig. 1. For cylindrical sinks (dislocations) the effective medium solutions are complex, and the sink strength is usually represented simply as

$$k_{i,d}^2 = \frac{2\pi\rho_d}{\log(1/r_{i,d}\sqrt{\pi\rho_d})} \quad (15)$$

instead with $r_{i,d}$ representing the dislocation capture radius for i .

Glasslike interstitial clusters are often expected to migrate only on one dimensional trajectories. The difference in behavior between one dimensionally and three dimensionally diffusing species is also easily handled in these models. The dimensionality of diffusion enters into the reaction rate coefficients k^+ . For 1d diffusion, the sink strengths are different, and can be written [43]

$$k_{ij}^2 = \frac{2D_i\sigma_{ij}C_j}{\lambda_i} \quad (16)$$

where σ_{ij} is the interaction cross section between the diffuser and the sink and λ_i is related to the total sink strength for i and defined

$$\lambda_i^{-1} = \sum_j \sigma_{ij} C_j. \quad (17)$$

The cross sections for spherical sinks are written as πr_{ij}^2 and $\rho_d r_{i,d}$ for spherical and dislocation sinks respectively. The interaction distances between one dimensional diffusers and dislocations $r_{i,d}$ have taken widely different values, with some work assuming that the clusters remain tightly bound to their glide cylinders and that the interactions distances are only a few Burgers vectors [44,45] and other models assuming relatively large cross sections as a consequence of strain interactions [27].

The reaction rate coefficients for sinks absorbing diffusers follow simply from the relationships with partial sink strengths as expressed in Eq. (13). Mutual interactions between diffusers can become more complex. In the case of the mutual interaction between two 3d diffusing species, the direct summation of the reaction rates for each species considered separately as the diffuser produces the correct rate. This can be implemented either as a single reaction with a rate given by

$$K_{ij}^+ = 4\pi r_{ij} (D_i + D_j) \quad (18)$$

or as two separate reactions each with its own rate

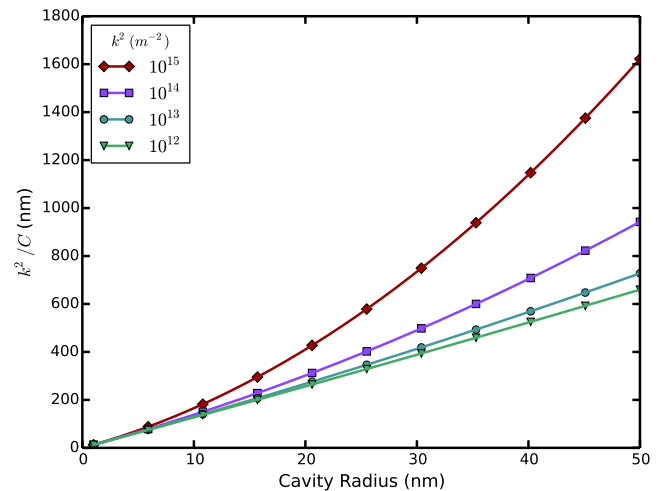


Fig. 1. The partial sink strength contribution of a single spherical defect absorber according to Eq. (14). The second order correction becomes relevant only at large radii, and only becomes dominant when the net system sink strength is rather high.

$$k_{ij}^+ = 4\pi r_{ij} D_i \quad (19)$$

$$k_{ji}^+ = 4\pi r_{ij} D_j \quad (20)$$

as a matter of preference, here assuming the higher order correction can be neglected. When one or both species diffuses one-dimensionally, the complexity arises. In many cases, the diffusivity of one species will be of significantly greater magnitude than the other, in which case the linear superposition of rates will create minimal error. In cases where the diffusivities are similar, a “master curve” approach has been developed to determine the appropriate rates of reaction [46,47]. In practice, the glide mobilities of one dimensionally diffusing clusters are quite high, rendering their concentration at any given time extraordinarily low and any mutual interactions between them negligible. The master curve also allows the consideration of rotations in glide direction, which may be significant for the smallest interstitial clusters. Sink strengths for anisotropic diffusion of other types have been developed [48], but apply only within certain limits and may not converge to correct sink strengths in isotropic or one dimensional limits.

Some clusters may be poorly represented as spherical sinks. This is particularly true of dislocation loops (e.g. sufficiently large interstitial clusters), which might be well described as spherical sinks up to no more than one or two nanometers in size. Past this point, the sink strengths are known to more closely resemble that of a network dislocation, with the details of the transition depending on how the sink strength is calculated. Seeger and Gosele found the sink strength of a dislocation loop with radius a and defect capture radius r to be [49]

$$k_L^2(a) = \frac{4\pi^2 a}{\log(8a/r)} \quad (21)$$

assuming $r < a$. Some care must be taken using this expression, as some means of avoiding the singularity at small loop radii must be used and it becomes invalid at sufficiently large loop sizes that the loop sink strength falls below the network dislocation sink strength of Eqs. (15) [50]. More recent studies have found that loop sink strengths are a rather complicated function of size and total sink density which can vary significantly with how the calculations are performed [51–53].

3.2.2. Capture sizes and biases

The morphology of a given cluster and the number of defects accumulated into it determine its size. Cavities, precipitates, and other clusters which can generally be described as spherical have radii

$$r_n = \left(\frac{3n\Omega}{4\pi} \right)^{1/3} \quad (22)$$

for a cluster with n members and a material with atomic volume Ω . Similarly, clusters better described as dislocation loops will have loop radii

$$r_n = \left(\frac{n\Omega}{\pi b} \right)^{1/2} \quad (23)$$

for a loop with Burgers vector magnitude b . The capture radius between two clusters may be written generically as

$$r_{ij} = r_i + r_j + r_{ij}^0 \quad (24)$$

where r^0 is the separation distance between the surfaces of the two clusters at which spontaneous combination occurs. At atomic resolution, there is likely to be a significant crystallographic effect, and as strictly defined, r^0 may vary significantly with the direction the defects are separated in the lattice. For our purposes it is sufficient to find an average quantity that produces the proper net rates of

capture. In practice this parameter has received little attention, and is simply fixed to a constant value (often zero) for all reactions.

The sink strengths and rate coefficients discussed above are derived in the absence of point defect interactions with the strain fields that may be generated by certain types of sink. The drift diffusion caused by these strain field interactions is chiefly responsible for the inequitable distribution of interstitial and vacancy type defects that drive many radiation induced microstructural changes. In the rate theory context, this presents itself as a change in sink strength for defects of different types, such that for a given sink j , the sink strength may differ between defects i . In such cases, it is convenient to write the sink strength in terms of a capture efficiency Z , such that $k_{ij}^2 = Z_{ij} C_j$ for point sinks or $k_{id}^2 = Z_{id} \rho_d$ for dislocations. In the absence of strain field interactions with the diffusing species, Z reduces to the expressions for sink strengths outlined previously. When Z differs between SIA and vacancy defects for a given sink, we may write

$$B_j = \frac{Z_{ij} - Z_{vj}}{Z_{vj}} \quad (25)$$

which is generally known as the bias in MFRT, and the definition may be presented with the SIA rather than vacancy capture efficiency in the denominator. Often, only the edge dislocation is considered to be a biased sink, with all others neutral. Several analytical expressions for the edge dislocation bias have been derived [54,55,6,56], with the simplest given by Miller [57]

$$B_d = \frac{\log |\Delta V_i / \Delta V_v|}{\log (1 / (r_{v,d} \sqrt{\pi \rho_d}))} \quad (26)$$

where $r_{v,d}$ is the dislocation capture radius for vacancies and ΔV_n is the relaxation volume of the point defect, a measure of the expansion or contraction of the system due to inserting the defect, which for most metals is 2 to 5 times larger in magnitude for the SIA than the vacancy. This low order approximation is sufficient for obtaining an estimate of the bias magnitude, and is plotted in Fig. 2. The biases in real materials may deviate significantly from expressions of this type due to anisotropy in the material (either elastic or diffusive), atomistic effects near the core, heterogeneous sink distributions, the interference of strain fields, and the segregation of solutes to sinks.

Conceptually, the differences in capture efficiency can be attributed to larger effective interaction radii between the defect and the

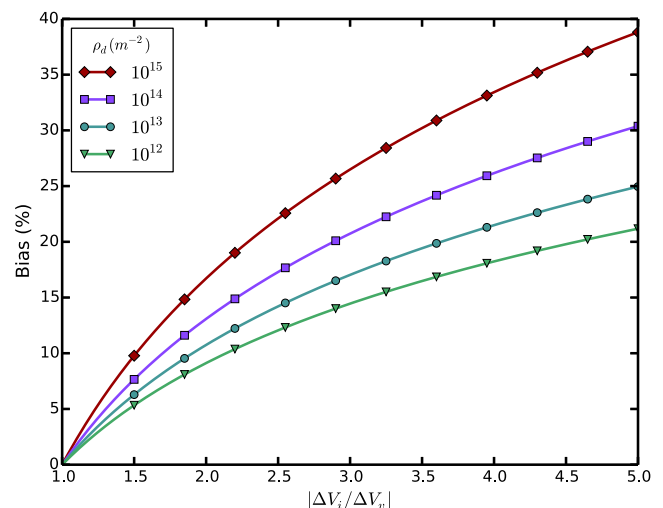


Fig. 2. Coarse estimates of the bias at various dislocation densities as a function of the ratio of the SIA and vacancy relaxation volume magnitudes as described by Eq. (26) when $r_{v,d} = 0.5$ nm.

sink. Heald [58] expresses the effective capture radius for a dislocation absorbing defect n as

$$r_{n,d} = b \left(1 + \frac{(1+\nu)\mu|\Delta V_n|}{6\pi(1-\nu)k_b T} \right) \quad (27)$$

where ν is the poisson ratio and μ is the shear modulus of the material. A similar concept applies to cavities, where the image force causes a slightly larger interaction distance for interstitials than vacancies at a magnitude roughly independent of the cavity size. Due to the primarily linear increase of cavity sink strength with radius, the resulting bias decreases with cavity size such that the bias of cavities is negligible at large sizes relevant to steady state void swelling, but may be quite significant in the cluster regime where voids are a few nanometers or smaller in diameter. Biases can be introduced in cluster dynamics calculations either by specifying interaction radii which account for the strain effect, or more commonly by applying an ex post facto increase to the appropriate sink strengths and reaction rate coefficients.

3.3. Dissociation

Dissociation of monomers from clusters is a thermally activated process that is of key importance to evolution, particularly at small cluster sizes. These events are represented in standard rate theory by subtracting the thermal concentration of the defect at the sink surface from its mean field value when determining a sink growth rate. The thermal concentration of i at the sink surface is written

$$C_{n,i}^0 = \frac{1}{\Omega} \exp \frac{-E_{n,i}^b}{k_b T} \quad (28)$$

where $E_{n,i}^b$ is the binding energy of i to n . The relationship between the defect capture and defect emission rates can be written in terms of the equilibrium conditions between the dissociation process and the reaction which reverses it

$$K_{i,n-i}^+ C_i^{eq} C_{n-i}^{eq} = K_{n,i}^- C_n^{eq} \quad (29)$$

where the equilibrium concentrations follow the law of mass action

$$\exp \frac{-E_{n,i}^b}{k_b T} = \frac{\Omega C_i^{eq} \Omega C_{n-i}^{eq}}{\Omega C_n^{eq}} \quad (30)$$

assuming that the free energy change associated with the process is dominated by E^b . The dissociation rate coefficient is then written in terms of its reaction pair as

$$k_{n,i}^- = \frac{K_{i,n-i}^+}{\Omega} \exp \frac{-E_{n,i}^b}{k_b T} \quad (31)$$

and the relationship between the reaction and dissociation process is identical to the relationship between the capture and thermal emission processes in older rate theory models Eq. (4). The cluster will absorb monomers more rapidly than emitting them whenever $C_i > 1/\Omega \exp -E_{n,i}^b/k_b T$, a critical monomer density past which the capture rate exceeds the emission rate. This density is plotted for selected binding energies as a function of temperature in Fig. 3

The binding energy is defined as the change in energy required to remove a defect from a cluster, and can be written as

$$E_{n,i}^b = E_i^f + E_{n-i}^f - E_n^f \quad (32)$$

the difference in formation energy E^f between the parent cluster and the sum of the two products. For cavities of radius r , the binding energy can be written in terms of the work done to change the cavity size against the force of the pressure p of any internal gas and the surface tension γ of the cavity. This is added to the formation

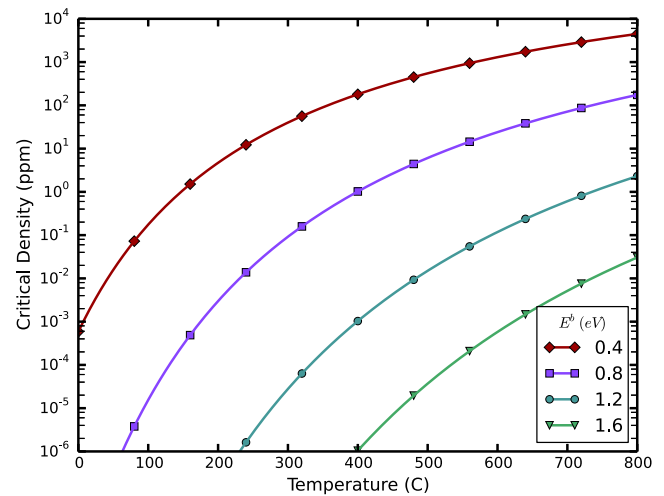


Fig. 3. The critical monomer density at which the rate of capture exceeds the rate of emission for selected values of the binding energy. For low binding energies and sufficiently high temperatures, the requisite density rapidly becomes unattainable.

energy E_i^f of the defect emitted from the parent cluster n to determine the total change in energy associated with the process as

$$E_{n,i}^b = E_i^f + \Delta n_i \Omega (p - 2\gamma/r(n)) \quad (33)$$

where Δn_i is the number of vacancies removed from the cavity as a result of the dissociation process (positive for vacancy emission and negative for SIA emission). The surface tension term is derived from the change in surface energy of the cavity

$$\frac{\partial E_\gamma}{\partial n} = \frac{\partial}{\partial n} (4\pi r(n)^2 \gamma) \quad (34)$$

for a cavity with n vacancies. In combination with the description of the radius of a spherical cluster Eq. (22), this reduces to $2\gamma\Omega/r$ in the continuum limit where n is large, but can also be written in terms of a unit change in the number of vacancies, such that

$$\Delta E = 4\pi\gamma \left(\frac{3\Omega}{4\pi} \right)^{2/3} (n^{2/3} - (n - \Delta n_i)^{2/3}) \quad (35)$$

for emission of a single vacancy. Capillarity law expressions of binding energies for voids,

$$E_{n,i}^b = C_0 + C_1 (n^{2/3} - (n - \Delta n_i)^{2/3}) \quad (36)$$

are based on this relationship, operate accurately down to quite small cluster sizes, and often perform well even for small SIA clusters despite their non-spherical geometry [59]. The parameters C_0 and C_1 are unique for each type of emitted defect i , and can be fit to atomistic data regarding binding energy or determined from the defect formation energies and the material surface tension. For larger clusters, alternative continuum expressions for binding energies can be derived by using formulae for the formation energetics of the desired defect structure, such as perfect or faulted loops [60].

In the standard rate theory models, usually only the thermal emission of vacancies was considered due to the much lower formation energy of the vacancy than the SIA in most materials. Generally, the SIA is also included as an emitable monomer in the cluster models. Though typically negligible from emission from pre-existing sinks thermally, SIA emission from over-pressurized gas bubbles is crucial to the nucleation of cavities in many materials. Cluster dynamics models of cavity growth typically include source terms for helium generation through (n,α) reactions or direct implantation. This creates a two dimensional cluster space

in which vacancy clusters absorb helium, and become over-pressurized. This significantly reduces the SIA binding energy, and at high enough helium to vacancy ratios promotes SIA emission to faster rates than vacancy emission. This phenomenon is the root of the “critical bubble” concept in cavity nucleation [61–63], which can now be modeled directly in the cluster dynamics framework. Athermal processes, such as the ballistic ejection of gas atoms from bubbles, can also be included through the incorporation of an appropriate constant frequency into the emission rate coefficients,

$$k_{n,i}^- = v_{n,i}^0 + \frac{K_{i,n-i}^+}{\Omega} \exp \frac{-E_{n,i}^b}{k_b T}. \quad (37)$$

4. Large scale simulations

As posed discretely in Eq. (6), the cluster dynamics method can describe most nucleation problems without difficulty. As the size of the clusters grow, the number of equations must grow as well. For instance, to describe the evolution of voids up to 50 nm in diameter, approximately 10^8 equations are required. This can be solved without difficulty if the reaction network remains sparse with few mobile species. As the network becomes denser, or an additional dimension becomes added to the problem due to the introduction of gas atoms, the problem quickly becomes intractable as posed. This “combinatorial explosion” of species and interactions renders most long term cluster growth problems untenable. For problems of intermediate scale, restriction of the phase space of evolution *a priori* to some region of interest (referred to as a phase cut) may be sufficient to obtain solutions at reasonable cost [64]. More generally, in order to evaluate these more complex problems the system must be re-written in order to reduce the number of active equations. After introducing the various approaches that have been taken to resolve such difficulties, we will apply each to a test problem, demonstrating their relative performance.

4.1. Stochastic integration

Rather than integrating the system directly through deterministic methods, the cluster model can be cast into a Monte Carlo formalism, and integrated stochastically [65]. This avoids the combinatorial explosion by effectively removing from consideration any regions of the cluster phase space which will not be physically sampled in the evolution trajectory of the system. The cost of the integration is then determined not by the number of possible cluster combinations, but rather the volume of the system considered. The concentration becomes quantized, with some integer number of each species found in the volume, and no need to store data for species where that number is zero. The reaction network of Eq. (7) becomes a list of reactions between the clusters found in the system at any given time as well as possible dissociations. To this event list is added any production rates as source terms and elimination events at distributed sinks.

The integration is then performed via the residence time algorithm [66]. That is, the probability of an event occurring at a given time step is governed by the ratio of its rate to the total rate of the system

$$p_i = \kappa_i / \sum_{i=1}^N \kappa_i \quad (38)$$

and the event for a step is selected by comparing a random number on the interval [0,1] to the cumulative probability distribution for the events available to the current system. The rates of these events

are taken directly from the formalism for the deterministic system as outlined in Table 1. Once an event is selected, the cluster counts are updated accordingly, and the event list is modified to reflect the new cluster content of the system. A second random number ξ is selected, and the time is incremented according to

$$\Delta t = - \frac{\sum_{i=1}^N \kappa_i}{\log(\xi)}. \quad (39)$$

This process is repeated until the desired time is reached.

One advantage of the stochastic approach is that it is relatively easy to add spatial dependence to the problem. This can be done simply by defining as many separate volumes as desired, and adding diffusion events between them to the rate lists as appropriate. This approach parallelizes easily, and with algorithms such as the synchronous parallel Monte Carlo technique [67], excellent scaling can be achieved, opening the door to very large scale spatially dependent simulations. The stochastic option is most attractive where the number of potential cluster configurations is very high. It is likely to outperform its deterministic counterparts for microstructure models coupling sink evolution to chemical segregation or precipitation in particular, where the cluster space may require two or more solute elements in addition to lattice defects. Another attractive feature of this approach is that the statistical nature of the method provides the ability to sample the distribution of possible microstructural outcomes from a particular set of physical processes, and provides some quantification of the range of possible experimental observations which are consistent with statistical fluctuation as opposed to indicating some meaningful difference in the underlying kinetic processes of the system.

The chief drawback to the stochastic methods is that performance is heavily influenced by the number of clusters in the system. A system with too many clusters in each volume generates long rate and species lists with long search or modification times, while a system with too few begins to misrepresent the proper physics of the system. These problems can usually be avoided with some prior knowledge about the behavior of the system as a whole, for instance if the nucleation density of a feature of interest is known. However, even with an optimal selection of the system volume, the step sizes in time remain small, and the terminal dose achievable by stochastic integrators remains relatively low for the family of MFRT based methods, on the order of a few dpa.

One means of softening this limitation is a hybrid scheme which handles the equilibrium between small clusters deterministically. These clusters typically have source terms, react rapidly amongst themselves and with sinks, and often have short lifetimes between dissociation events. By removing many of these events from the stochastic system, the residence time for stochastic steps can be greatly increased for some systems. The remainder of the phase space can still be evolved stochastically, with source terms in the stochastic regime which are determined by the concentrations and reaction rates of the clusters in the deterministic regime,

Table 1

Expressions for the rates at which events occur based on the number N of clusters in a volume and the mesh spacing h .

Event	Execution	Rate
Reaction	Remove i and j , add $i+j$	$k_{ij}^+ N_i N_j / h^3$
Dissociation	Remove n , and $n-i$ and i	$k_{n,i}^- N_n$
Source	Add n	$g_n h^3$
Sink	Remove n	$D_n k_n^2 N_n$
Transport	Remove n from one node, add n to an adjacent node	$D_n N_n / h^2$

$$g_n^{\text{interface}} = \sum_{i+j=n} k_{ij}^+ C_i C_j \quad (40)$$

and the reaction rates between a stochastic object and a deterministic concentration require scaling by the system volume (e.g. C_i becomes $N_i = C_i h^3$) when generating the event list as in Table 1.

4.2. Grouping schemes

One alternative to render the large scale problem tractable for deterministic methods in complex cluster spaces is the grouping scheme. The basic concept is to use a single equation to represent the concentration of a “group” of clusters with similar properties, which might require dozens, hundreds, or many millions of equations in the discrete approach. The challenge in developing a successful grouping scheme is generating a set of rate coefficients for the reduced system which accurately reproduce the physics of the discrete system. Early attempts to develop a grouping scheme [68] have later been proved unsuitable, with the primary flaw being a broadening of the size distribution [69]. More recently, a scheme has been developed based on conserving the moments of the size distribution [70]. In this approach, each group is represented by a pair of equations, one describing the mean density of the group and another describing its first moment. Additional equations for the first moment must be supplied for each additional dimension in the cluster phase space (e.g. if gas atoms are included). Though developed for monomer fluxes, this scheme can be generalized to include interactions between arbitrarily sized cluster groups in multiple dimensions with all groups capable of interaction regardless of size, with higher order interpolations of the size distribution offering similar performance and accuracy [71].

The disadvantage of grouping schemes is that all of the methods which preserve the size distribution properties allow for negative values of the concentration to develop. This is an issue related to the Gibbs and Runge phenomenon, where oscillations develop around sharp transitions in a function using Fourier series or polynomial interpolation approximations. For grouping, these oscillations are of low magnitude compared to the peak values of the distribution in most cases, and do not substantially influence the behavior of the system. Where the size distribution experiences sharp transitions, the oscillations can become pronounced, allowing the development of unphysical effects. Thus, as with the stochastic integrators, some knowledge of the behavior of the system is required *a priori*, such that any regions of the phase space which will develop a sharp transition are handled using small group widths relative to the length of the transitions, or even with discrete equations if necessary. Regions of the phase space which contain such sharp transitions, for instance the helium to vacancy ratio at which trap mutation occurs above which cluster densities rapidly drop to zero, are often identifiable in advance and the group widths can be set accordingly in practice.

4.3. Fokker-Planck

A similar approach to eliminating the need for representing each cluster density discretely is to cast the problem in the Fokker-Planck form [72–77]. In this approach, one represents the cluster distribution as a continuous function of cluster size x , and the evolution of the distribution is described as a drift diffusion problem,

$$\frac{\partial C(x, t)}{\partial t} = \frac{\partial}{\partial x} \left(f(x, t) C(x, t) - \frac{\partial d(x, t) C(x, t)}{\partial x} \right) \quad (41)$$

where $f(x, t)$ describes the drift force to grow or shrink clusters of size x and $d(x, t)$ describes the diffusion force to spread the size dis-

tribution at x . These coefficients can be constructed from the coefficients native to discrete cluster dynamics by summing over all mobile clusters j

$$f(x) = \sum_j j \left(k_{j,x}^+ C_j - k_{x,j}^- \right) \quad (42)$$

$$d(x) = \frac{1}{2} \sum_j j^2 \left(k_{j,x}^+ C_j + k_{x,j}^- \right) \quad (43)$$

With this, the drift term can be positive or negative but the diffusion term is always positive, and promotes spreading of any cluster distribution regardless of the details of the underlying kinetics. These equations exist on a continuum, while the formalism of cluster dynamics involves a discrete space. Generally, the problem is separated into a discrete regime which treats the species which are mobile and/or introduced as primary damage, and a continuous regime of predominantly immobile species evolved with Fokker-Planck. A transitional or overlap regime may be required at the interface between the continuous and discrete regions to obtain an accurate solution. Additionally, this approach limits the number of mobile clusters, and regions of the distribution described with a Fokker-Planck evolution scheme will not be capable of mutual interactions such as coalescence. With these considerations, Fokker-Planck appears to be the most difficult of the techniques to apply in a general fashion.

These methods were introduced in the radiation effects context for Frenkel pair damage problems, and were used to treat size distribution evolution before the full cluster dynamics methods became popular [8]. More recent models have included the effects of additional mobile clusters and source terms. As with grouping schemes, the Fokker-Planck approach can be applied to higher dimensional spaces. Sharp transitions in the size distribution also pose a problem for these methods, which tend to over-broaden the distribution in such cases, and may become unstable, though the exact response depends on the discretization scheme used to solve the equations in the continuous regime [76]. The Fokker-Planck form of the system can also be integrated stochastically, which may avoid some of these issues [73]. As with the grouping methods, some prior knowledge of the evolution trajectory of the system can be used to help select the scheme used, or the discretization size required to obtain a sufficiently accurate solution. As with the stochastic techniques, parallel computing can be leveraged to efficiently pursue spatially resolved simulations for any of the deterministic approaches.

4.4. Demonstration case

To illustrate the application of a few of these techniques in practice, we present a physically simplified (but still potentially computationally expensive) problem which involves the formation of

Table 2

The parameters used for a simple example problem, designed to illustrate the performance of various CD integration techniques.

T	575	K
$g(i_1, v_1)$	10^{-3}	dpa/s
Ω	0.0118	nm ³
$r_0(\text{all})$	0.37	nm
$D_0(i_1, v_1)$	10^{11}	nm ² /s
$E_{i_1}^m$	0.34	eV
$E_{v_1}^m$	0.67	eV
$D(v \geq 2)$	0	nm ² /s
$E_{v_n}^b$	$1.73 - 2.59(n^{2/3} - (n-1)^{2/3})$	eV
ρ_d	5×10^{13}	m ⁻²
B_d	5%	

voids under Frenkel pair damage conditions. The parameters used in the simulations are chosen to mimic bcc Fe, and are given in Table 2. This test is posed without spatial dependence, interstitial cluster evolution, or the mobility of large species, rendering one of the simplest possible scenarios a cluster dynamics method might be expected to analyze. Nonetheless, the voids that form can reach quite large sizes, requiring on the order of 10^7 equations in the discrete CD solution. A comparison of the size distributions generated by the discrete, grouped, Fokker-Planck, and stochastic approaches are shown in Fig. 4, along with the void density and average size as a function of dose. We note here that this problem is particularly disadvantageous to the stochastic method, which gains a comparative advantage only on significantly more complicated problems which involve many mobile clusters, many dimensions in the phase space, or spatial resolution over large, complex domains. A hybrid stochastic method where the SIA and the smallest three vacancy clusters are handled deterministically was also tested, and its performance is demonstrated as well.

The computational expense of each mode of simulation depends on implementation details, how well the solver is optimized for a particular problem, error tolerances and discretization widths in the deterministic schemes, and the simulation volume considered in stochastic schemes. We report the total integration times listed in Table 3 for each of the methods as applied to this problem, noting that detailed attention to these factors may allow

Table 3

The total integration time for each of the CD approaches, to obtain the results plotted in Fig. 4. For the stochastic methods, average times are given.

Approach	Final dose (dpa)	Time (s)
Discrete	100	2.2×10^4
Fokker-Planck	100	2.4
Grouped	100	0.3
Stochastic	0.1	4.9×10^4
Hybrid Stochastic	100	7.2×10^4

faster solutions than we have achieved. The Fokker-Planck simulations were run on a non-uniform mesh using a central difference discretization, which has the potential disadvantage of inducing oscillations in the solution, as is visible in the size distribution at the highest doses. Alternative discretization schemes can avoid such instabilities, but generally introduce numerical diffusion that can compromise the accuracy of the size distribution and disturb the overall properties of the cluster population, such as mean size. The grouping schemes allow more rapid changes in the discretization width, and accordingly require fewer active equations, particularly at early doses. Both deterministic approaches provide a significant reduction in runtime.

The stochastic integrations were performed in an $8 \times 10^6 \text{ nm}^3$ volume, and halted at 0.1 dpa. The lowest density visible in such

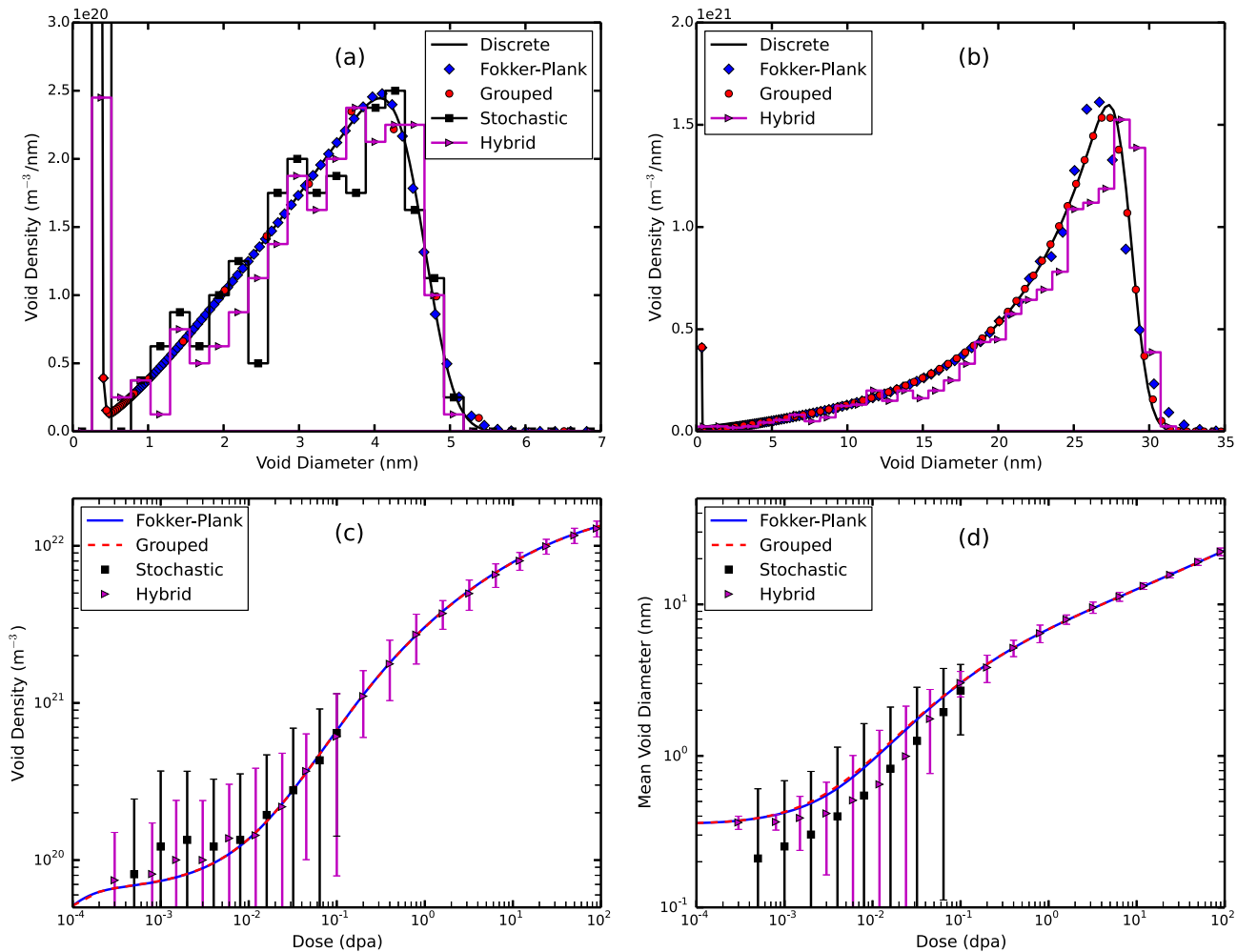


Fig. 4. Properties of the void distribution as simulated with discrete CD and the Fokker-Planck, grouped, stochastic, and hybrid stochastic variants. Compared properties include (a) the size distribution at 0.1 dpa, (b) the size distribution at 100 dpa, (c) the void density as a function of dose and (d) the average void diameter as a function of dose. For stochastic simulations, 40 iterations were run, and the variation in the net density and mean size is also shown.

a system is slightly larger than 10^{20} m^{-3} , and at early doses when the expected cluster density is near or below this level, some of the stochastic integrators deviate to some extent from the discrete solution. The frequency of defect production, elimination at sinks, and formation and subsequent dissociation of small vacancy clusters is high compared to the evolution of the void population, and treating the equations for the small clusters deterministically significantly reduced the number of steps required. Accordingly, the hybrid scheme demonstrated a three order of magnitude decrease in computational expense, and allowed the stochastic integrator to reach the full dose. Both of the stochastic approaches produce some variability in the cluster distribution, and an interval representing two standard deviations on values for density and size is also shown.

5. Applications

CD models have, in essence, two varieties of application, which can be thought of as forward and inverse problems. The first mode of operation is fundamentally predictive. In this approach, a CD model is constructed from the available data for a given material with the aim of replicating the evolution of its microstructure. The objective here is to predict radiation induced changes in physical or mechanical properties, such as swelling or hardening, connecting knowledge on the atomic scale to behaviors on the microstructural scale, and becomes a key link in multi-scale modeling framework for the theory of radiation damage. The agreement, or lack thereof, between experiments and CD models parameterized with the available atomistic information can lend confidence to the existing data on a particular material, or point to gaps in understanding which require further investigation.

In the second, the model is applied in direct connection with a specific set of experiments with the goal of inferring information about the material system. In this inverse problem approach, specific migration energies or binding energies might be deduced by repeated simulation with the aim of reproducing a particular experimental result. One example is the analysis of resistivity recovery experiments. Interpretation of resistivity recovery experiments in the past focused on estimating the migration energies of the monomer SIA and vacancy, while assigning any other recovery stages was vague and highly speculative. With the advent of cluster dynamics, the recovery spectrum for a given array of binding and migration energies can be generated directly, and the various stages can be tied directly to the responsible mechanisms [78,79]. Further, the energetics of the various defect migration and dissociation processes can often be deduced by ensuring the spectrum generated by a model is consistent with experiment.

5.1. Noble gas bubbles

As a consequence of the almost universally negligible solubility of He in metals, He bubbles can be generated in both plasma facing components -in which He is continuously injected into the component- and in materials exposed to neutrons -in which (n,α) reactions lead to He generation in the bulk. In the latter, the gas bubbles may serve as nuclei for the formation of cavities, which would otherwise be unable to overcome the nucleation barrier imposed by thermal dissociation of small vacancy clusters. In each case, the study of gas bubble formation can lead to a more comprehensive understanding of the microstructures which ultimately develop, and provide insight into the factors that make for radiation resistant materials in these environments. This is accomplished with a two dimensional cluster phase space incorporating helium and vacancies, which describes the entire

range of possible gas pressures within any size distribution of cavities, illustrated in Fig. 5.

Such models have been applied to the formation of small bubbles, examining the transition from over-pressurized bubbles which grow by the absorption of helium to thermally stable under-pressurized cavities which grow through the excess vacancy flux induced by the dislocation bias for interstitials [80,75,81,76,77]. The binding energy of vacancies to small clusters is lower due the curvature dependence of the surface tension contributions in Eq. (33). Consequently there is a critical size, below which the thermal emission of vacancies proceeds more rapidly than arrival of excess vacancies from the bias driven supersaturation. In this thermal evaporation regime, helium is required to stabilize the cluster against vacancy emission, and any clusters in this regime will shrink due to vacancy emission until reaching an equilibrium He to vacancy ratio where the gas pressure is sufficient to preclude further evaporation. Long term growth of these stable bubbles can only proceed by the absorption of additional helium, to allow the inclusion of further vacancies without entering the evaporation regime. This process continues until reaching the critical size, at which point no additional helium is necessary, and the cluster acquires vacancies more rapidly than releasing them regardless of its gas content, a sequence which forms the basis of the critical bubble concept from traditional rate theory arguments [61–63]. Cluster dynamics can function as a high fidelity implementation of the critical bubble model, taking into account atomistic data regarding the binding energies of particular helium vacancy clusters below sizes where equation of state based arguments are meaningful, as well as providing full size and pressure distributions for the bubble population in both the nucleation and growth stages.

In the context of plasma surface interactions, cluster dynamics models been used to investigate the fate of helium injected into surface layers of materials such as tungsten [82,83,38]. In these applications the bubble nucleation is considered to occur primarily as a consequence of helium aggregation and trap mutation, rather than the interaction of helium with radiation induced defect clusters. In this environment, the flux of helium is much higher relative to the point defect supersaturation, and the trajectory of cluster evolution is restricted to much higher He to vacancy ratios. However, the pressure increases from repeated He absorption events, and eventually the bubble must grow in size by emitting an SIA

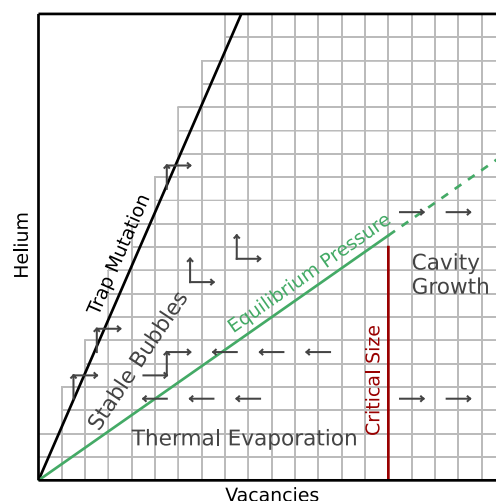


Fig. 5. A schematic of the phase space involved in bubble formation processes. Regimes where HeV clusters shrink due to thermal emission of vacancies, grow in a vacancy supersaturation, or grow due to trap mutation are shown, along with examples of evolution trajectories.

or cluster of SIAs to relax the internal pressure of the gas. In this context, models have been compared both to MD simulations and to implantation experiments in order to investigate the kinetics of helium aggregation. Further, these models allow an analysis of the effect of the different timescales between MD and experimental implantation rates, which is several orders of magnitude, and aid in interpreting the reliability of MD results in this light.

Others have used comparisons to thermal desorption spectroscopy (TDS) measurements to investigate the binding energetics of small helium clusters and bubbles [36,37]. During TDS, helium is injected into a material which is subsequently heated. The rate of helium desorption from the sample is continuously monitored during heating, giving some indication of the temperatures at which critical dissociation processes are activated. TDS experiments can be reproduced with high fidelity in a spatially resolved model, and significant information about the energetics of helium dissociation processes can be deduced by selecting energies to fit the desorption spectrum in the model to the ones measured experimentally.

5.2. Dislocation loops and black spot damage

Rate theory based models are also applied to the nucleation and growth of interstitial clusters. Here, direct comparisons between *in situ* ion irradiation experiments and CD models have produced some valuable insights into point defect cluster properties and behaviors [84]. At lower temperatures, *in situ* irradiations of metals generally feature “black spot” damage, consisting of a very high density of small damage features on the order of a few nanometers in size. While in fcc metals these are generally considered to result from the collapse of individual cascades, in the bcc systems the characteristics of the defect accumulation are very different [85], and MD simulations of cascades do not indicate that visibly sized clusters can be formed directly in isolated events, except in close proximity to a surface. The role of the CD model is to investigate the space of defect kinetics that allows such damage to form in these system, with an eye to matching not just density and size evolution, but also additional features such as depth distribution through the foil or the temperature dependence of the damage state [35,33,45]. Similar models can also be applied to bulk neutron irradiation conditions [86,87]. In general, it has been found across bcc systems that the long range mobility of small dislocation loops (crowdion bundles) must be significantly lower than the predictions of MD, attributed to trapping interactions between these clusters and impurities in the system. However, even a small fraction of the primary damage created in the form of sessile clusters rapidly leads to a over-nucleation, and models predict complete saturation of the microstructure with immobile species within fractions of a dpa.

Larger dislocation loops, which develop at higher temperatures or at longer irradiation times, have also been examined [88,50,89,44,53]. In these cases, capturing the transition in sink strength between defect clusters and extended loops can be crucial to an accurate representation of nucleation and growth kinetics. Further, when biases are applied to the dislocation network structure, they must also be applied to the loops, in a manner that may depend non-trivially on loop size. The bias specified can have a significant impact on the predictions of the model, particularly as pertaining to long term growth, and competing estimates for the bias factors of loops have been advanced [53,52]. Investigations of loop sink strengths and bias factors are consistent primarily in predicting complex (often non-monotonic) functions of loop size and net sink strength, with the bias of small loops usually exceeding the network at low sink density, and falling below it at high sink densities. Further, the anisotropy of defect diffusion may play a significant role in loop development, particularly in cases where the

distribution of line directions is different between the loop population and the network, or between two distinct loop populations. For instance, CD models in the hcp structure have examined the effects of anisotropic diffusion, both of monomer defects and of crowdion bundles, in an attempt to better understand the preferential growth of vacancy and interstitial type loops on different crystallographic planes [89,44].

Even small changes to the biases of loops and assumptions about the primary damage state can significantly alter the behavior of a loop nucleation and growth model. As a simple demonstration of this, consider a simple Frenkel pair damage scenario, with all the clusters except the single SIA and vacancy held immobile, and neglecting the presence of vacancy clusters. As a surrogate for the possible deviation of the loop biases from the network sink strength we introduce a loop bias function

$$B_L = B_d \left(1 - \frac{\epsilon}{r_L} \right) \quad (44)$$

where the default case of $\epsilon = 0$ returns a loop bias equivalent to the network bias at all sizes. Positive or negative values for ϵ will render a favorable or unfavorable bias for loop growth at small sizes, while restoring the network bias at large loop radii. Under default bias conditions, we will also consider a small fraction of the mass being produced in cascades as diverted into sessile interstitial clusters of size 20. For the simulation parameters in Table 2, the loop evolution under several conditions is shown in Fig. 6.

The changes in bias which are applied here are not large, but already have a substantial impact on loop development. The impact on density is greater than an order of magnitude, and an even slightly unfavorable bias condition is sufficient to eliminate any long term growth. Similarly, even low frequency production of sessile clusters is capable of significantly increasing the nucleation densities of loops, and at 1% of the cascade mass partitioned to sessile interstitial clusters, the loop densities already become unphysically high. Additional changes in behavior are introduced when the long range mobility and dimensionality of small interstitial clusters are considered - which are likely impacted by interactions with solutes - and experiments have shown that the density of loops that develop can be strongly affected by impurity content.

5.3. Precipitation

The CD formalism has also been used to analyze radiation enhanced precipitation [90–96]. The diffusion of substitutional solutes in metals is typically dominated by the vacancy mechanism, in which the diffusion of solutes progresses by the exchange of solutes with vacant sites in the lattice. Accordingly, the magnitude of the diffusion coefficient of the solute itself increases with the vacancy density, as the probability of a neighboring site being vacant increases. Thus, for the thermal diffusion of solute n following an Arrhenius relation

$$D_n^{\text{th}} = D_n^0 \exp \frac{-E_n^a}{k_b T} \quad (45)$$

the total activation energy E_n^a is comprised of both the energy required to form vacancies in the lattice and the barriers involved in the atomic level exchange processes. The thermal diffusion coefficient might be equivalently written in terms of the thermal vacancy density as,

$$D_n^{\text{th}} = D_{n,v}^0 \Omega C_v^{\text{th}} \exp \frac{-(E_n^a - E_v^f)}{k_b T} \quad (46)$$

where the energy $E_n^a - E_v^f$ constitutes an effective barrier for the vacancy mediated migration of the solute, and is similar in form to the diffusion coefficient for lattice defects. In an irradiation envi-

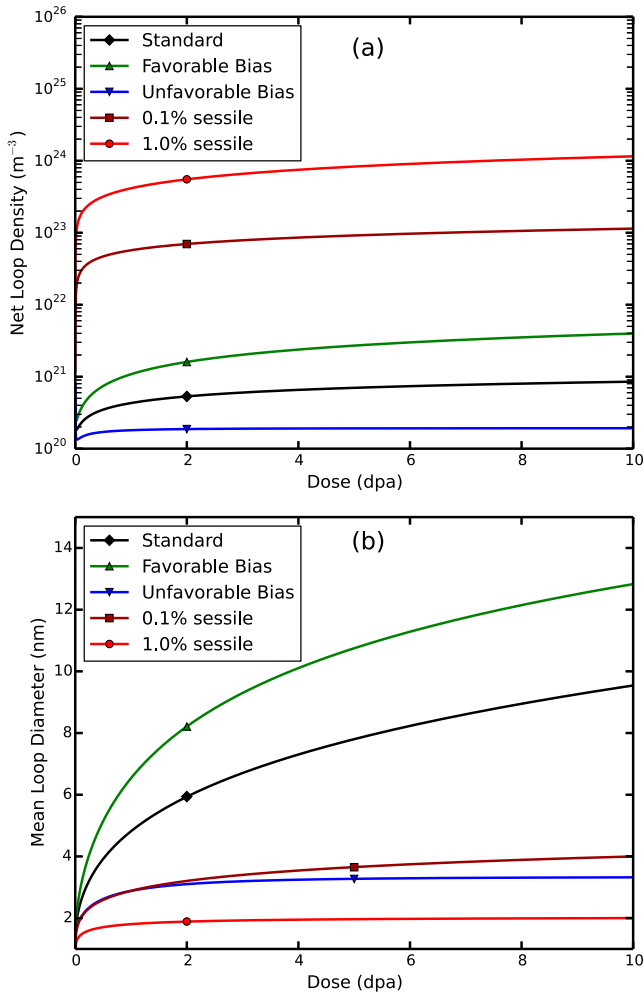


Fig. 6. The effect of a favorable ($\epsilon = 0.1$ nm) or unfavorable ($\epsilon = -0.1$ nm) bias for the development of interstitial type loops as compared to the standard condition ($\epsilon = 0$). The effect of producing sessile interstitial clusters in cascades is also demonstrated. The (a) net density and (b) mean diameter of interstitial clusters is shown over the first 10 dpa.

ronment, the probability of finding a neighboring vacant site increases due to the production of point defects in damage events. The increase in solute diffusivity by the vacancy mechanism can be written by multiplying the thermal diffusivity by the vacancy supersaturation

$$D_n^{\text{irr}} = D_n^0 \frac{C_v}{C_v^{\text{th}}} \exp \frac{-E_n^a}{k_b T} = D_n^0 C_v \exp \frac{-(E_n^a - E_v^f)}{k_b T}, \quad (47)$$

which may evolve with time. This produces, in general, a diffusivity and set of reaction rate coefficients which can vary over the course of irradiation exposure, or take constant values in a non-irradiation environment.

The focus of the cluster evolution is on the formation of solute clusters and their evolution into extended precipitates. The same type of reaction network is built, based on the migration of isolated solutes to solute clusters of various sizes. The diffusion coefficient of the solute itself is a function of the vacancy density, which can be determined by the solution of point defect balance equations for the system in question. This can be done by assuming a simple MFRT model considering only Frenkel pair damage, recombination, and loss at sinks, but it is possible to incorporate a full CD representation of the point defect population as well, and evolve both systems simultaneously. Small solute clusters may also migrate

as a consequence of exchange of the surface atoms with vacancies, and these effects can be incorporated easily into the cluster dynamics models [94], and additional modes of self-diffusion might be activated under irradiation as well, for instance from the presence of self-interstitials and vacancy cluster defects. Individual solutes may dissociate from precipitates, analogous to the loss of point defects from clusters. In this case, the binding energy is governed by the interfacial energy between the precipitate and the bulk, rather than the surface tension and internal pressure as was the case for cavities.

5.4. Reduced order models and connection with chemico-mechanical problems

The analysis of data generated from CD models can provide numerically tractable routes towards integration of microstructure evolution in the context of constitutive model development. Indeed, a relatively large body of work, relying on atomistic simulations, discrete defect dynamics and crystal plasticity constitutive modeling have proposed approaches to predict the effects of irradiation induced defects on strength and hardening of metals. A prime example of success in the case of hierarchical modeling lies in the calibration of superposition laws (e.g. the dispersed barrier hardening model), used in the context of crystal plasticity, that allow predictions of the simultaneous effects of voids and SIA loops on mechanical strength [97,98]. Particular emphasis has been placed on the case of α Iron. As such, modeling of the mechanical response of out of pile metals can be achieved; that is from the knowledge of the post irradiation microstructure, mechanical strength can be predicted. Ideally though, both CD and constitutive models could be coupled, similarly to OKMC/discrete dislocation dynamics coupling originally introduced by Ghoniem et al., in order to predict in-pile mechanical response of the material. There are two aspects of the coupling between these scales that can be considered. A full coupling between CD and constitutive laws (CL) requires not only that the effects of irradiation induced defects on strength and plasticity are taken into account, but that the effects of internal stresses and plastically induced defects (i.e. dislocations) are considered in the evolution of the radiation induced defect populations. In recent years, some encouraging work has been proposed in this direction, notably leveraging DFT and atomistic simulations to compute interaction dipole tensors allowing to predict drift effects on diffusion due to strains [99,100]. We note that the timescales associated with plastic deformation and with diffusive processes are vastly different, rendering the natural form of the problem ill posed and complicating the process of efficient time integration. In this regard, the use of a full CD/CL coupling may become computationally prohibitive.

One notes however, that at least in the case of partial coupling between CL and CD approaches, CD models can be used to parameterize a reduced order model (ROM). In developing the latter, the typical questions that arise pertain to both the mathematical form of the ROM and its calibration. MFRT can supply the mathematical form of the ROM, and CD can be used to calibrate effective diffusivities and sink strengths. As an example, we present the case of He implantation in thin films, for which a full CD model was used to predict He concentration profiles in an α Fe thin film as a function of thickness and dose [101]. The effective diffusivity for helium was deduced from its aggregate retention in the foil, obtained from the full CD simulation. A reduced order form of the problem can be expressed as

$$\frac{\partial C_{\text{He}}}{\partial t} = \nabla \cdot (D_{\text{eff}} \cdot \nabla C_{\text{He}}) + g_{\text{He}} \quad (48)$$

where D_{eff} corresponds to the effective diffusion coefficient for helium within the system as a whole. This has a particular solution

$$C_{\text{He}}(x, t) = \frac{g_{\text{He}}(Lx - x^2)}{2D_{\text{eff}}} - \frac{4g_{\text{He}}}{D_{\text{eff}}L} \left(\sum_{n \text{ odd}} \left(\frac{L}{n\pi} \right)^3 \sin \left(\frac{n\pi x}{L} \right) \exp \left(-\frac{n^2 \pi^2 D_{\text{eff}} t}{L^2} \right) \right) \quad (49)$$

in a foil of width L . The appropriate effective diffusivity causes the concentration profiles generated by this ROM and the full CD model to match, and depends on temperature and dose rate as shown in Fig. 7. One notes that D_{eff} varies by six order of magnitudes. Importantly the difference between the effective He diffusivities and the *a priori* simpler choice of using the He interstitial diffusivity is at least four orders of magnitude.

Additionally, CD can also be used to address one of the key questions pertaining to the use of self-ion implantations as proxies for neutron irradiation conditions. One important motivation for using ions as a surrogate damage source is that the dose rate is three to five orders of magnitude higher than those corresponding to in service neutron flux conditions. This benefit is also a drawback, as it can create an imbalance between the kinetics of point defect generation and diffusion. In compensation, the diffusion kinetics are often accelerated by increasing the exposure temperature in order to obtain the same damage state, with expressions estimating the proper magnitude of this temperature shift derived from MFRT [7]. Using CD, one can address these issues with higher fidelity by sampling the dpa, dpa rate and temperature spaces over all irradiation conditions relevant to both self-ion and neutron implantation conditions. From the resulting dataset detailing the irradiated microstructure (i.e. cluster density, distributions, etc.), one can find equivalent irradiation conditions leading to the same microstructure state. A map detailing the predicted necessary temperature shift to reach equivalent damaged microstructures with distinct dpa/rates is shown below as presented in reference [102]. CD has been applied to investigate the temperature shift in cases of cascade implantation of pure Fe as illustrated in Fig. 8, fully parameterized up to doses of 0.01 dpa [102] as well as pure Mo [103].

6. Limitations

6.1. Multiscale interfaces and degrees of freedom

For each cluster in the system, the size, shape, mobility, and reaction network constituents must be specified. For each reaction

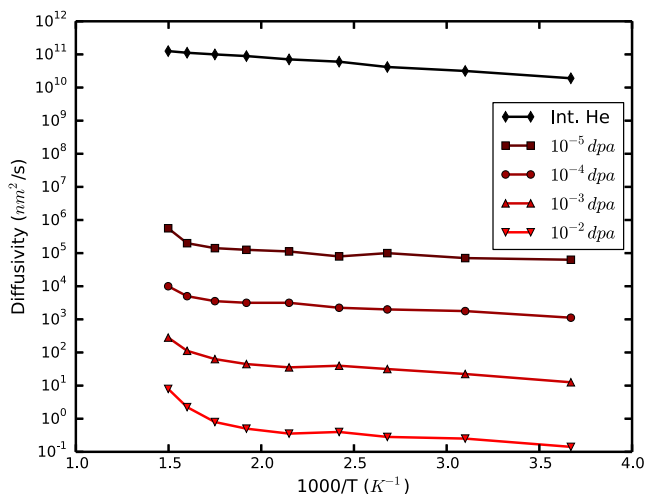


Fig. 7. The effective diffusivity of Helium in α -Fe [101], used for generating reduced order models as discussed in the main text.

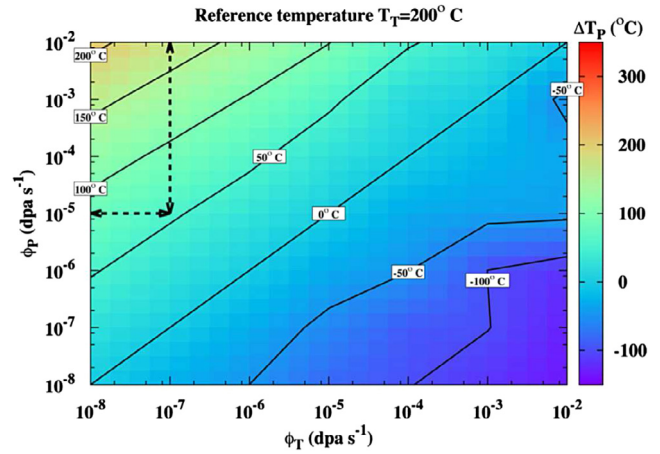


Fig. 8. Temperature shift required to emulate target dose rate ϕ_T with a proxy dose rate ϕ_p , as calculated using CD at a target temperature of 200 °C [102].

network constituent, the rate coefficient must be determined, which contains some dependence on the properties of the reacting or dissociating species, but may have other contributing factors specific to the interaction in question, such as a bias or binding energy. Consequently, the number of parameters required to describe a system can grow rapidly. A migration energy and diffusion prefactor need to be supplied for each mobile species, and a binding energy for every dissociation pathway. Interaction radii between each pair of clusters capable of reacting must be specified, and the form for the rate coefficient appropriate to the geometry must be set. Typically, discrete values are only considered for a small handful of clusters, and the vast majority are described by continuous expressions that are functions of only a few parameters, such as the capillarity laws for the binding energies as in Eq. (36). Regardless, there are typically a minimum of ten parameters associated with even the simplest, coarsest models and there is generally some uncertainty associated with each.

Further, in the optimal case in which the energetics of point defect clusters can be supplied with confidence for pure materials from atomistic models, the parameters relevant to engineering alloys of practical interest may be far different. Consider the case of ferritic steels, where there is some evidence that the initial phase of cascade evolution is not significantly affected by the presence of Cr in the matrix [104,105], but that the long range migration of point defects may be modified much more significantly [106,107]. In such circumstances one must either use the defect mobility from the pure system and assume the alloying effects are negligible, or apply an effective diffusivity usually inferred in some manner from experiments. Similar questions arise regarding bias factors, cluster diffusion dimensionality, long range mobility of larger clusters, and even the cascade damage state itself. The propagation of uncertainties in the inputs to a given model's predictions are rarely addressed comprehensively, in part due to the complex, multidimensional nature of any such analysis.

6.2. Validity of mean field assumptions

Where damage is produced by cascades, the initial sub-nanosecond stage is believed to be well described by MD simulations. Later timescales in cascade aging are not available to conventional MD techniques but the damage remains highly spatially correlated, in violation of the assumptions pertaining to systems uniformity which support the mean-field reaction rate formula. Some additional intra-cascade recombination and self-clustering is expected, therefore, prior to reaching mean field conditions.

Indeed, object kinetic Monte Carlo simulations of damage evolution show little difference from mean field methods when the damage is introduced without spatial correlation, but do indicate some degree of difference in behavior when the damage is introduced in the form of cascades [108–110]. However, with stochastic integration, adaptive spatial meshes can be used which account for the insertion of a cascade directly [87].

When deterministic integration is used, a more subtle means of bridging the suppressed timescales must be applied. Often this comes from aging cascades directly with Monte Carlo methods [111], where defects which escape a particular aging volume surrounding the cascade or survive to the end of the anneal are considered to have reached mean field conditions, and those which recombine during the aging process are not. The results of such a technique may depend on the size of the annealing cell used to generate the aged cluster distributions or the amount of time elapsed during annealing. In such cases, volumes chosen to represent the net system sink strength provide the most accurate description of cluster distribution which reaches the mean field [112]. Consequently, the source terms applicable to mean field methods may change spatially and over time.

Length scales can also be suppressed in these models, and the mean field formulation conceals any spatial dependence that the defect evolution may have around sinks which are not explicitly represented in the mesh. The sink strength is, in reality, impacted by the proximity of various sinks to one another, and a configuration of tightly clustered sinks can be less effective at absorbing defects than the same density of sinks distributed more regularly. Further, the reaction rates between mobile defects are determined from the mean field densities, which are higher than the spatially dependent concentrations near sinks and lower further away from them. The rates of mutual interaction between mobile defect species, such as recombination, are overpredicted near sinks and underpredicted far away from them. For instance, one might write the average error in the recombination rate due to spatial correlations between defect concentration profiles as

$$\epsilon_{i,v} = \frac{1}{V} \int_V k_{i,v}^+ [C_i C_v - (C_i + \Delta c_i(\vec{r}))(C_i + \Delta c_v(\vec{r}))] dV \quad (50)$$

where $\Delta c(\vec{r})$ indicates the local deviation from the mean field concentration at point $\vec{r}$, such that the volume average of Δc is zero. The error then simplifies to

$$\epsilon_{i,v} = -\frac{1}{V} \int_V k_{i,v}^+ \Delta c_i(\vec{r}) \Delta c_v(\vec{r}) dV \quad (51)$$

which is independent of the mean field values themselves.

The degree to which cluster evolution in the mean field may differ from the spatially resolved case due to spatial correlation and other effects is generally poorly quantified, and remains a particularly intriguing area of investigation for CD models. Errors will appear for all mutual interactions, not just recombination, and has the potential to distort net rates of homogeneous nucleation of vacancy and interstitial clusters. For defects with the same shape of concentration profile, such as self-clustering of two of the same type of defect, the error is always negative indicating that nucleation is underpredicted. The effect may be particularly exacerbated in the vicinity of sinks with complex strain fields, where the deviations of the local point defect concentrations vary significantly from defect to defect, and processes forbidden in the mean field may become available or vice versa. The magnitude and long term effects of these types of errors are difficult to estimate *a priori* as the shape of each defect profile depends on its interaction with the sink in question as well as other defects. One example of the

improved ability of modern models to capture the effect of profile shapes is the appearance of a “rabbit ears” vacancy profile in spatially resolved calculations of defect evolution in thin films or grains [113,34]. This occurs under recombination dominant conditions, where the vacancy supersaturation actually increases near sinks as a consequence of the local depletion of the more mobile interstitial species, resulting in a spike in vacancy concentration at the grain boundary surface. Consequently, the nucleation of vacancy clusters is enhanced in these regions while the nucleation of interstitial clusters is diminished or even suppressed altogether. This highlights the importance of spatial resolution in rate theory based methods, and motivates the development of accurate methods to treat as many sinks as possible explicitly (as local defect absorbers) rather than implicitly as a homogenized lossy medium.

7. Summary

Starting from the pioneering development of mean field rate theory, and leveraging the advances in computational physics to both reveal physical phenomena driving microstructure evolution during irradiation and calibrate laws pertaining to these unit processes, several numerical and fundamental developments to the cluster dynamics framework have been proposed over the past few decades. In this manuscript, we have highlighted the connection between the current CD modeling and the traditional MFRT approach, illustrating both the new capabilities offered and the fundamental similarity in the underlying physical processes considered. Approaches to allow for a full treatment of defect cluster distributions have been proposed including grouping schemes, the Fokker-Planck method, and stochastic solvers. The underlying techniques, limitations and advantages of these methods have been presented in this manuscript. In particular, the potential limitations of the mean field formalism for accurately representing the capture rates of point defects at sinks and the probability of point defect interactions due to spatial correlations were examined. The progress in developing spatially resolved CD models, which may relax some of these concerns, has also been addressed.

Further, the manuscript provides an overview of the typical applications of CD models, including advances in predicting loop and cavity nucleation, and modeling the formation of gas bubbles and precipitates. The advantages that the additional complexity these modern techniques bring - such as the higher fidelity representation of size distributions and primary damage states - has been discussed in connection with the difficulty they present with regard to parametric uncertainty in complex material systems and scale bridging from atomistic methods. Some of complexities are inherited from MFRT predecessors, while others are freshly introduced as a consequence of the larger space of physical information required for a full CD model. Additionally, future directions for the use of CD have been proposed, notably towards developing reduced order models that can be used in conjunction with other techniques such as crystal plasticity constitutive modeling. As microstructures for materials to be used for nuclear energy applications become more complex, the need for further connecting the processed microstructure, its history, and its complex chemistry in understanding radiation tolerance has become increasingly necessary. To this effect spatially resolved methods may allow high fidelity simulations of complex microstructures, and the reduction in cost achievable through techniques described in this document may permit coupling of the micro-chemical and point-defect transport aspects of microstructural evolution, providing a route towards improving the predictive capabilities of radiation damage modeling.

Acknowledgements

The authors wish to acknowledge the full support of the Los Alamos National Laboratory LDRD office under contract 20170615ER

Appendix A. Sink strengths in the effective medium

The mean field expressions for sink strengths are derived by solving for the flux into a particular sink, and relating this to the current that flows into other similar sinks. This involves making certain symmetry assumptions about the system surrounding the sink, and these have varied from author to author. The most rigorously constructed method of determining sink strengths was advanced by Brailsford and Bullough [41], who described a system with defects flowing into a central sink which is embedded in a lossy medium which absorbs defects at a rate consistent with the total system sink strength. In this approach, they argue, the random nature of the surrounding sink structure is preserved and represented self-consistently through the action of the lossy medium. In this construction, the central sink may or may not be surrounded by a sink-free region, and the remainder of the sinks are smeared out into a lossy medium in the ensemble average, as illustrated in Fig. A.9.

Within the sink free region, the concentration profile of a defect n at steady state is governed by

$$0 = \kappa_n - \nabla \cdot J_n \quad (\text{A.1})$$

with κ_n the net rate of production of n by all possible means. Here, we will make a distinction between κ and the rate of production in damage events g . In the homogenized medium the governing equation contains a loss term

$$0 = \kappa_n - \nabla \cdot J_n - D_n k_n^2 c_n \quad (\text{A.2})$$

where k_n^2 refers to the *total* system sink strength, and the spatially dependent concentration c_n is distinct from the mean field concentration C_n . The boundary conditions are that the flux vanishes far from the sink, such that

$$0 = \kappa_n - D_n k_n^2 c_n(\infty) \quad (\text{A.3})$$

and the concentration far from the sink, therefore, is identical to the mean field concentration. Additionally, the concentration vanishes at the interaction distance between the sink and the defect r_0

$$0 = c(r_0). \quad (\text{A.4})$$

The current into a given set of sinks j is computed by finding the flux based on the concentration profile around the sink, taking the closed integral at the sink surface, and scaling the result by the density of that sink C_j ,

$$I_{n,j} = -D_n C_j \oint_S \nabla c_n \cdot dA_S. \quad (\text{A.5})$$

This current must be equivalent to the one described by the homogenized partial sink strength for j , such that

$$\frac{I_{n,j}}{D_n C_n} = k_{ij}^2 = -\frac{C_j}{C_n} \oint_S \nabla c_n \cdot dA_S. \quad (\text{A.6})$$

We note here that in the MFRT context, the inner boundary was often maintained at the thermal equilibrium concentration instead to accurately capture the net flux, ultimately resulting in what is analogous to a dissociation term in the cluster dynamics formalism. Here, we are only interested in the inward flux, and the outward flux is handled separately by the means described in Section 3.3.

The solution depends on the geometry of the sink, the interaction energy landscape, the dimensionality of the defect diffusion, and the size of the sink free region (if any) considered. For illustrative purposes, we demonstrate the spherical sink with isotropic defect diffusion and no sink free region. In this case, the entire problem is described by

$$0 = \kappa_n + D_n \nabla^2 c_n - D_n k_n^2 c_n \quad (\text{A.7})$$

in spherical coordinates which produces a concentration profile which is determined by the boundary conditions and independent of κ_n

$$c(r) = C_n \left(1 - \frac{r_0}{r} \exp k_n(r_0 - r) \right) \quad (\text{A.8})$$

with gradient of the form

$$\frac{\partial c(r)}{\partial r} = C_n \left(\frac{r_0}{r^2} + \frac{k_n r_0}{r} \right) \exp k_n(r_0 - r). \quad (\text{A.9})$$

When inserted into Eq. (A.6) this produces a sink strength

$$k_{nj}^2 = 4\pi r_0^2 C_j (1/r_0 + k_n) \quad (\text{A.10})$$

which is equivalent to Eq. (14) in the main text.

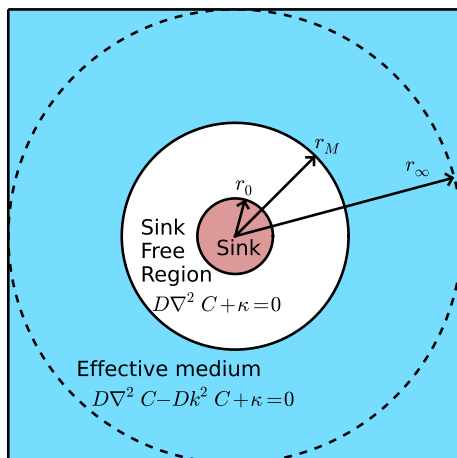


Fig. A.9. An illustration of a central sink, with surrounding sink free region and effective lossy medium.

References

- [1] V. Sears, Kinetics of void growth in irradiated metals, *J. Nucl. Mater.* 39 (1) (1971) 18–26, [https://doi.org/10.1016/0022-3115\(71\)90179-6](https://doi.org/10.1016/0022-3115(71)90179-6). <<http://www.sciencedirect.com/science/article/pii/0022311571901796>>.
- [2] J.L. Katz, H. Wiedersich, Nucleation of voids in materials supersaturated with vacancies and interstitials, *J. Chem. Phys.* 55 (3) (1971) 1414–1425, <https://doi.org/10.1063/1.1676236>.
- [3] H. Wiedersich, On the theory of void formation during irradiation, *Radiat. Effects* 12 (1–2) (1972) 111–125, <https://doi.org/10.1080/00337577208231128>.
- [4] A. Brailsford, R. Bullough, The rate theory of swelling due to void growth in irradiated metals, *J. Nucl. Mater.* 44 (2) (1972) 121–135, [https://doi.org/10.1016/0022-3115\(72\)90091-8](https://doi.org/10.1016/0022-3115(72)90091-8). <<http://www.sciencedirect.com/science/article/pii/0022311572900918>>.
- [5] A. Brailsford, R. Bullough, M. Hayns, Point defect sink strengths and void swelling, *J. Nucl. Mater.* 60 (3) (1976) 246–256, [https://doi.org/10.1016/0022-3115\(76\)90139-2](https://doi.org/10.1016/0022-3115(76)90139-2). <<http://www.sciencedirect.com/science/article/pii/0022311576901392>>.
- [6] P. Heald, M. Speight, Point defect behaviour in irradiated materials, *Acta Metall.* 23 (11) (1975) 1389–1399, [https://doi.org/10.1016/0001-6160\(75\)90148-0](https://doi.org/10.1016/0001-6160(75)90148-0). <<http://www.sciencedirect.com/science/article/pii/0001616075901480>>.
- [7] L. Mansur, Correlation of neutron and heavy-ion damage, *J. Nucl. Mater.* 78 (1) (1978) 156–160, [https://doi.org/10.1016/0022-3115\(78\)90514-7](https://doi.org/10.1016/0022-3115(78)90514-7). <<http://www.sciencedirect.com/science/article/pii/0022311578905147>>.

- [8] M.F. Wehner, W.G. Wolfer, Vacancy cluster evolution in metals under irradiation, *Philos. Mag. A* 52 (2) (1985) 189–205, <https://doi.org/10.1080/01418618508237618>.
- [9] H. Wiedersich, P. Okamoto, N. Lam, A theory of radiation-induced segregation in concentrated alloys, *J. Nucl. Mater.* 83 (1) (1979) 98–108, [https://doi.org/10.1016/0022-3115\(79\)90596-8](https://doi.org/10.1016/0022-3115(79)90596-8). <http://www.sciencedirect.com/science/article/pii/0022311579905968>.
- [10] K. Russell, Phase stability under irradiation, *Prog. Mater. Sci.* 28 (3) (1984) 229–434, [https://doi.org/10.1016/0079-6425\(84\)90001-X](https://doi.org/10.1016/0079-6425(84)90001-X). <http://www.sciencedirect.com/science/article/pii/007964258490001X>.
- [11] P.T. Heald, M.V. Speight, Steady-state irradiation creep, *Phil. Mag.* 29 (5) (1974) 1075–1080, <https://doi.org/10.1080/14786437408226592>.
- [12] E. Savino, C. Tom, Irradiation creep by stress-induced preferential attraction due to anisotropic diffusion (SIPA-AD), *J. Nucl. Mater.* 108 (Supplement C) (1982) 405–416, [https://doi.org/10.1016/0022-3115\(82\)90509-8](https://doi.org/10.1016/0022-3115(82)90509-8). <http://www.sciencedirect.com/science/article/pii/0022311582905098>.
- [13] J. Matthews, M. Finnis, Irradiation creep models an overview, *J. Nucl. Mater.* 159 (Supplement C) (1988) 257–285, [https://doi.org/10.1016/0022-3115\(88\)90097-9](https://doi.org/10.1016/0022-3115(88)90097-9). <http://www.sciencedirect.com/science/article/pii/0022311588900979>.
- [14] L. Mansur, Theory and experimental background on dimensional changes in irradiated alloys, *J. Nucl. Mater.* 216 (1994) 97–123, [https://doi.org/10.1016/0022-3115\(94\)90009-4](https://doi.org/10.1016/0022-3115(94)90009-4). <http://www.sciencedirect.com/science/article/pii/0022311594900094>.
- [15] T. Okita, W. Wolfer, A critical test of the classical rate theory for void swelling, *J. Nucl. Mater.* 327 (2) (2004) 130–139, <https://doi.org/10.1016/j.jnucmat.2004.01.026>. <http://www.sciencedirect.com/science/article/pii/S0022311504000765>.
- [16] A.J.E. Foreman, W.J. Phythian, C.A. English, The molecular dynamics simulation of irradiation damage cascades in copper using a many-body potential, *Philos. Mag. A* 66 (5) (1992) 671–695, <https://doi.org/10.1080/01418619208201584>.
- [17] A. Calder, D. Bacon, A molecular dynamics study of displacement cascades in iron, *J. Nucl. Mater.* 207 (Supplement C) (1993) 25–45, [https://doi.org/10.1016/0022-3115\(93\)90245-T](https://doi.org/10.1016/0022-3115(93)90245-T). <http://www.sciencedirect.com/science/article/pii/002231159390245T>.
- [18] D.J. Bacon, T.D. de la Rubia, Molecular dynamics computer simulations of displacement cascades in metals, *J. Nucl. Mater.* 216 (Supplement C) (1994) 275–290, [https://doi.org/10.1016/0022-3115\(94\)90016-7](https://doi.org/10.1016/0022-3115(94)90016-7). <http://www.sciencedirect.com/science/article/pii/0022311594900167>.
- [19] W. Phythian, R. Stoller, A. Foreman, A. Calder, D. Bacon, A comparison of displacement cascades in copper and iron by molecular dynamics and its application to microstructural evolution, *J. Nucl. Mater.* 223 (3) (1995) 245–261, [https://doi.org/10.1016/0022-3115\(95\)00022-4](https://doi.org/10.1016/0022-3115(95)00022-4). <http://www.sciencedirect.com/science/article/pii/0022311595000224>.
- [20] R.E. Stoller, Point defect survival and clustering fractions obtained from molecular dynamics simulations of high energy cascades, *J. Nucl. Mater.* 233 (Part 2) (1996) 999–1003, [https://doi.org/10.1016/S0022-3115\(96\)00261-9](https://doi.org/10.1016/S0022-3115(96)00261-9). <http://www.sciencedirect.com/science/article/pii/S0022311596002619>.
- [21] R. Stoller, G. Odette, B. Wirth, Primary damage formation in bcc iron, *J. Nucl. Mater.* 251 (Supplement C) (1997) 49–60, [https://doi.org/10.1016/S0022-3115\(97\)00256-0](https://doi.org/10.1016/S0022-3115(97)00256-0), proceedings of the International Workshop on Defect Production, Accumulation and Materials Performance in an Irradiation Environment <http://www.sciencedirect.com/science/article/pii/S0022311597002560>.
- [22] P. Heald, M. Speight, The influence of cascade damage on irradiation creep and swelling, *J. Nucl. Mater.* 64 (1) (1977) 139–144, [https://doi.org/10.1016/0022-3115\(77\)90017-4](https://doi.org/10.1016/0022-3115(77)90017-4). <http://www.sciencedirect.com/science/article/pii/0022311577900174>.
- [23] B. Wirth, G. Odette, D. Maroudas, G. Lucas, Energetics of formation and migration of self-interstitials and self-interstitial clusters in iron, *J. Nucl. Mater.* 244 (3) (1997) 185–194, [https://doi.org/10.1016/S0022-3115\(96\)00736-2](https://doi.org/10.1016/S0022-3115(96)00736-2), radiation Materials Science in Technology Applications <http://www.sciencedirect.com/science/article/pii/S0022311596007362>.
- [24] Y. Osetsky, D. Bacon, A. Serra, B. Singh, S. Golubov, Stability and mobility of defect clusters and dislocation loops in metals, *J. Nucl. Mater.* 276 (1) (2000) 65–77, [https://doi.org/10.1016/S0022-3115\(99\)00170-1](https://doi.org/10.1016/S0022-3115(99)00170-1). <http://www.sciencedirect.com/science/article/pii/S0022311599001701>.
- [25] Y.N. Osetsky, D.J. Bacon, A. Serra, B.N. Singh, S.I. Golubov, One-dimensional atomic transport by clusters of self-interstitial atoms in iron and copper, *Phil. Mag.* 83 (1) (2003) 61–91, <https://doi.org/10.1080/0141861021000016793>.
- [26] C.H. Woo, B.N. Singh, The concept of production bias and its possible role in defect accumulation under cascade damage conditions, *Phys. Status Solidi (b)* 159 (2) (1990) 609–616, <https://doi.org/10.1002/psb.2221590210>.
- [27] H. Trinkaus, B. Singh, A. Foreman, Glide of interstitial loops produced under cascade damage conditions: possible effects on void formation, *J. Nucl. Mater.* 199 (1) (1992) 1–5, [https://doi.org/10.1016/0022-3115\(92\)90433-L](https://doi.org/10.1016/0022-3115(92)90433-L). <http://www.sciencedirect.com/science/article/pii/002231159290433L>.
- [28] B.N. Singh, A.J.E. Foreman, Production bias and void swelling in the transient regime under cascade damage conditions, *Philos. Mag. A* 66 (6) (1992) 975–990, <https://doi.org/10.1080/01418619208248002>.
- [29] C.H. Woo, B.N. Singh, Production bias due to clustering of point defects in irradiation-induced cascades, *Philos. Mag. A* 65 (4) (1992) 889–912, <https://doi.org/10.1080/01418619208205596>.
- [30] H. Trinkaus, B. Singh, A. Foreman, Impact of glissile interstitial loop production in cascades on defect accumulation in the transient, *J. Nucl. Mater.* 206 (2) (1993) 200–211, [https://doi.org/10.1016/0022-3115\(93\)90124-H](https://doi.org/10.1016/0022-3115(93)90124-H). <http://www.sciencedirect.com/science/article/pii/002231159390124H>.
- [31] B. Singh, H. Trinkaus, C. Woo, Production bias and cluster annihilation: why necessary?, *J. Nucl. Mater.* 212 (Part 1) (1994) 168–174, [https://doi.org/10.1016/0022-3115\(94\)90050-7](https://doi.org/10.1016/0022-3115(94)90050-7). <http://www.sciencedirect.com/science/article/pii/0022311594900507>.
- [32] A. Dunn, R. Dingreville, E. Martinez, L. Capolungo, Synchronous parallel spatially resolved stochastic cluster dynamics, *Comput. Mater. Sci.* 120 (2016) 43–52, <https://doi.org/10.1016/j.commatsci.2016.04.013>. <http://www.sciencedirect.com/science/article/pii/S0927025616301720>.
- [33] A.A. Kohnert, B.D. Wirth, Cluster dynamics models of irradiation damage accumulation in ferritic iron. I. Trap mediated interstitial cluster diffusion, *J. Appl. Phys.* 117 (15) (2015) 154305, <https://doi.org/10.1063/1.4918315>.
- [34] A.Y. Dunn, L. Capolungo, E. Martinez, M. Cherkaoui, Spatially resolved stochastic cluster dynamics for radiation damage evolution in nanostructured metals, *J. Nucl. Mater.* 443 (1–3) (2013) 128–139, <https://doi.org/10.1016/j.jnucmat.2013.07.009>. <http://www.sciencedirect.com/science/article/pii/S0022311513008878>.
- [35] D. Xu, B.D. Wirth, M. Li, M.A. Kirk, Combining in situ transmission electron microscopy irradiation experiments with cluster dynamics modeling to study nanoscale defect agglomeration in structural materials, *Acta Mater.* 60 (10) (2012) 4286–4302, <https://doi.org/10.1016/j.actamat.2012.03.055>. <http://www.sciencedirect.com/science/article/pii/S1359645412002455>.
- [36] D. Xu, B.D. Wirth, Modeling spatially dependent kinetics of helium desorption in bcc iron following he ion implantation, *J. Nucl. Mater.* 403 (1) (2010) 184–190, <https://doi.org/10.1016/j.jnucmat.2010.06.025>. <http://www.sciencedirect.com/science/article/pii/S0022311510002667>.
- [37] X. Hu, D. Xu, B.D. Wirth, Quantifying he-point defect interactions in fe through coordinated experimental and modeling studies of he-ion implanted single-crystal fe, *J. Nucl. Mater.* 442 (1–3, Supplement 1) (2013) S649–S654, <https://doi.org/10.1016/j.jnucmat.2013.02.064>. <http://www.sciencedirect.com/science/article/pii/S0022311513004704>.
- [38] S. Blondel, K.D. Hammond, L. Hu, D. Maroudas, B.D. Wirth, Modeling helium segregation to the surfaces of plasma-exposed tungsten as a function of temperature and surface orientation, *Fusion Sci. Technol.* 71 (1) (2017) 22–35, <https://doi.org/10.13182/FST16-112>. <http://www.tandfonline.com/doi/abs/10.13182/FST16-112>.
- [39] A. Dunn, R. Dingreville, E. Martinez, L. Capolungo, Identification of dominant damage accumulation processes at grain boundaries during irradiation in nanocrystalline α -Fe: a statistical study, *Acta Mater.* 110 (2016) 306–323, <https://doi.org/10.1016/j.actamat.2016.03.026>. <http://www.sciencedirect.com/science/article/pii/S1359645416301793>.
- [40] B.P. Uberuaga, L.J. Vernon, E. Martinez, A.F. Voter, The relationship between grain boundary structure, defect mobility and grain boundary sink efficiency, *Sci. Rep.* 5 (1) (2015), <https://doi.org/10.1038/srep09095>.
- [41] A.D. Brailsford, R. Bullough, The theory of sink strengths, *Philos. Trans. Roy. Soc. Lond. Ser. A Math. Phys. Sci.* 302 (1465) (1981) 87–137. <http://www.jstor.org/stable/36884>.
- [42] M.V. Smoluchowski, *Z. phys. Chem.* 92 (1917) 192.
- [43] V. Borodin, Rate theory for one-dimensional diffusion, *Physica A* 260 (3) (1998) 467–478, [https://doi.org/10.1016/S0378-4371\(98\)00338-0](https://doi.org/10.1016/S0378-4371(98)00338-0). <http://www.sciencedirect.com/science/article/pii/S0378437198003380>.
- [44] A. Barashev, S. Golubov, R. Stoller, Theoretical investigation of microstructure evolution and deformation of zirconium under neutron irradiation, *J. Nucl. Mater.* 461 (2015) 85–94, <https://doi.org/10.1016/j.jnucmat.2015.02.001>. <http://www.sciencedirect.com/science/article/pii/S0022311515000859>.
- [45] A.A. Kohnert, B.D. Wirth, Cluster dynamics models of irradiation damage accumulation in ferritic iron. II. Effects of reaction dimensionality, *J. Appl. Phys.* 117 (15) (2015) 154306, <https://doi.org/10.1063/1.4918316>.
- [46] H. Heinisch, B. Singh, S. Golubov, The effects of one-dimensional glide on the reaction kinetics of interstitial clusters, *J. Nucl. Mater.* 283 (Part 2) (2000) 737–740, [https://doi.org/10.1016/S0022-3115\(00\)00258-0](https://doi.org/10.1016/S0022-3115(00)00258-0), 9th Int. Conf. on Fusion Reactor Materials <http://www.sciencedirect.com/science/article/pii/S0022311500002580>.
- [47] H. Trinkaus, H.L. Heinisch, A.V. Barashev, S.I. Golubov, B.N. Singh, 1d to 3d diffusion-reaction kinetics of defects in crystals, *Phys. Rev. B* 66 (2002) 060105, <https://doi.org/10.1103/PhysRevB.66.060105>. <https://link.aps.org/doi/10.1103/PhysRevB.66.060105>.
- [48] C. Woo, Theory of irradiation deformation in non-cubic metals: effects of anisotropic diffusion, *J. Nucl. Mater.* 159 (Supplement C) (1988) 237–256, [https://doi.org/10.1016/0022-3115\(88\)90096-7](https://doi.org/10.1016/0022-3115(88)90096-7). <http://www.sciencedirect.com/science/article/pii/0022311588900967>.
- [49] A. Seeger, U. Gsele, Steady-state diffusion of point defects to dislocation loops, *Phys. Lett. A* 61 (6) (1977) 423–425, [https://doi.org/10.1016/0375-9601\(77\)90355-3](https://doi.org/10.1016/0375-9601(77)90355-3). <http://www.sciencedirect.com/science/article/pii/0375960177903553>.
- [50] F. Christien, A. Barbu, Effect of self-interstitial diffusion anisotropy in electron-irradiated zirconium: a cluster dynamics modeling, *J. Nucl. Mater.* 346 (2) (2005) 272–281, <https://doi.org/10.1016/j.jnucmat.2005.06.024>. <http://www.sciencedirect.com/science/article/pii/S0022311505003107>.
- [51] C. Woo, The sink strength of a dislocation loop in the effective medium approximation, *J. Nucl. Mater.* 98 (3) (1981) 279–294, [https://doi.org/10.1016/0022-3115\(81\)90154-9](https://doi.org/10.1016/0022-3115(81)90154-9). <http://www.sciencedirect.com/science/article/pii/0022311581901549>.
- [52] V. Dubinko, A. Abyzov, A. Turkin, Numerical evaluation of the dislocation loop bias, *J. Nucl. Mater.* 336 (1) (2005) 11–21, <https://doi.org/10.1016/j.jnucmat.2005.06.024>.

- jnuclmat.2004.07.034. <<http://www.sciencedirect.com/science/article/pii/S0022311504006178>>.
- [53] T. Jourdan, Influence of dislocation and dislocation loop biases on microstructures simulated by rate equation cluster dynamics, *J. Nucl. Mater.* 467 (2015) 286–301, <https://doi.org/10.1016/j.jnuclmat.2015.09.046>. <<http://www.sciencedirect.com/science/article/pii/S002231151530235X>>.
 - [54] W.G. Wolfer, M. Ashkin, Diffusion of vacancies and interstitials to edge dislocations, *J. Appl. Phys.* 47 (3) (1976) 791–800, <https://doi.org/10.1063/1.322710>.
 - [55] H. Rauh, D. Simon, On the diffusion process of point defects in the stress field of edge dislocations, *Phys. Status Solidi (a)* 46 (2) (1978) 499–510, <https://doi.org/10.1002/pssa.2210460213>.
 - [56] W.G. Wolfer, The dislocation bias, *J. Comput. Aided Mater. Des.* 14 (3) (2007) 403–417, <https://doi.org/10.1007/s10820-007-9051-3>.
 - [57] K. Miller, Dislocation bias and point-defect relaxation volumes, *J. Nucl. Mater.* 84 (1) (1979) 167–172, [https://doi.org/10.1016/0022-3115\(79\)90160-0](https://doi.org/10.1016/0022-3115(79)90160-0). <<http://www.sciencedirect.com/science/article/pii/0022311579901600>>.
 - [58] P.T. Heald, The preferential trapping of interstitials at dislocations, *Phil. Mag.* 31 (3) (1975) 551–558, <https://doi.org/10.1080/01418617508226537>.
 - [59] N. Soneda, T.D. de la Rubia, Defect, production, annealing kinetics and damage evolution in α -Fe: an atomic-scale computer simulation, *Philos. Mag. A* 78 (5) (1998) 995–1019, <https://doi.org/10.1080/01418619808239970>.
 - [60] S.J. Zinkle, L.E. Seitzman, W.G. Wolfer, I. energy calculations for pure metals, *Philos. Mag. A* 55 (1) (1987) 111–125, <https://doi.org/10.1080/01418618708209803>.
 - [61] R. Stoller, G. Odette, Analytical solutions for helium bubble and critical radius parameters using a hard sphere equation of state, *J. Nucl. Mater.* 131 (2) (1985) 118–125, [https://doi.org/10.1016/0022-3115\(85\)90450-7](https://doi.org/10.1016/0022-3115(85)90450-7). <<http://www.sciencedirect.com/science/article/pii/0022311585904507>>.
 - [62] M. Hayns, M. Wood, R. Bullough, A theoretical evaluation of dual-beam irradiation experiments, *J. Nucl. Mater.* 75 (2) (1978) 241–250, [https://doi.org/10.1016/0022-3115\(78\)90005-3](https://doi.org/10.1016/0022-3115(78)90005-3). <<http://www.sciencedirect.com/science/article/pii/0022311578900053>>.
 - [63] L. Mansur, W. Coghan, Mechanisms of helium interaction with radiation effects in metals and alloys: a review, *J. Nucl. Mater.* 119 (1) (1983) 1–25, [https://doi.org/10.1016/0022-3115\(83\)90047-8](https://doi.org/10.1016/0022-3115(83)90047-8). <<http://www.sciencedirect.com/science/article/pii/0022311583900478>>.
 - [64] D. Xu, X. Hu, B.D. Wirth, A phase-cut method for multi-species kinetics: sample application to nanoscale defect cluster evolution in alpha iron following helium ion implantation, *Appl. Phys. Lett.* 102 (1) (2013) 011904, <https://doi.org/10.1063/1.4773876>.
 - [65] J. Marian, V.V. Bulatov, Stochastic cluster dynamics method for simulations of multispecies irradiation damage accumulation, *J. Nucl. Mater.* 415 (1) (2011) 84–95, <https://doi.org/10.1016/j.jnuclmat.2011.05.045>. <<http://www.sciencedirect.com/science/article/pii/S002231151100523X>>.
 - [66] A. Bortz, M. Kalos, J. Lebowitz, A new algorithm for Monte Carlo simulation of Ising spin systems, *J. Comput. Phys.* 17 (1) (1975) 10–18, [https://doi.org/10.1016/0021-9991\(75\)90060-1](https://doi.org/10.1016/0021-9991(75)90060-1). <<http://www.sciencedirect.com/science/article/pii/0021999175900601>>.
 - [67] E. Martinez, J. Marian, M. Kalos, J. Perlado, Synchronous parallel kinetic Monte Carlo for continuum diffusion-reaction systems, *J. Comput. Phys.* 227 (8) (2008) 3804–3823, <https://doi.org/10.1016/j.jcp.2007.11.045>. <<http://www.sciencedirect.com/science/article/pii/S002199910700527X>>.
 - [68] M. Kiritani, Analysis of the clustering process of supersaturated lattice vacancies, *J. Phys. Soc. Jpn.* 35 (1) (1973) 95–107, <https://doi.org/10.1143/JPSJ.35.95>.
 - [69] M. Koiwa, On the validity of the grouping method comments on analysis of the clustering process of supersaturated lattice vacancies, *J. Phys. Soc. Jpn.* 37 (6) (1974) 1532–1536, <https://doi.org/10.1143/JPSJ.37.1532>.
 - [70] S.I. Golubov, A.M. Ovcharenko, A.V. Barashev, B.N. Singh, Grouping method for the approximate solution of a kinetic equation describing the evolution of point-defect clusters, *Philos. Mag. A* 81 (3) (2001) 643–658, <https://doi.org/10.1080/01418610108212164>.
 - [71] A.A. Kohnert, B.D. Wirth, Grouping techniques for large-scale cluster dynamics simulations of reaction diffusion processes, *Modell. Simul. Mater. Sci. Eng.* 25 (1) (2017) 015008. <<http://stacks.iop.org/0965-0393/25/i=1/a=015008>>.
 - [72] M.F. Wehner, W.G. Wolfer, Numerical evaluation of path-integral solutions to Fokker-Planck equations, *Phys. Rev. A* 27 (1983) 2663–2670, <https://doi.org/10.1103/PhysRevA.27.2663>. <<https://link.aps.org/doi/10.1103/PhysRevA.27.2663>>.
 - [73] M. Surh, J. Sturgeon, W. Wolfer, Master equation and Fokker-Planck methods for void nucleation and growth in irradiation swelling, *J. Nucl. Mater.* 325 (1) (2004) 44–52, <https://doi.org/10.1016/j.jnuclmat.2003.10.013>. <<http://www.sciencedirect.com/science/article/pii/S0022311503004847>>.
 - [74] M.P. Surh, J. Sturgeon, W. Wolfer, Vacancy cluster evolution and swelling in irradiated 316 stainless steel, *J. Nucl. Mater.* 328 (2) (2004) 107–114, <https://doi.org/10.1016/j.jnuclmat.2004.03.005>. <<http://www.sciencedirect.com/science/article/pii/S0022311504001114>>.
 - [75] M.P. Surh, J.B. Sturgeon, W.G. Wolfer, Void nucleation, growth, and coalescence in irradiated metals, *J. Nucl. Mater.* 378 (1) (2008) 86–97, <https://doi.org/10.1016/j.jnuclmat.2008.05.009>. <<http://www.sciencedirect.com/science/article/pii/S0022311508002596>>.
 - [76] T. Jourdan, G. Bencteux, C. Adjanor, Efficient simulation of kinetics of radiation induced defects: a cluster dynamics approach, *J. Nucl. Mater.* 444 (1) (2014) 298–313, <https://doi.org/10.1016/j.jnuclmat.2013.10.009>. <<http://www.sciencedirect.com/science/article/pii/S0022311513011483>>.
 - [77] D. Brimbal, A. Fournier, A. Barbu, Cluster dynamics modeling of the effect of high dose irradiation and helium on the microstructure of austenitic stainless steels, *J. Nucl. Mater.* 468 (2016) 124–139, <https://doi.org/10.1016/j.jnuclmat.2015.11.007>. <<http://www.sciencedirect.com/science/article/pii/S0022311515303159>>.
 - [78] J.D. Torre, C.-C. Fu, F. Willaime, A. Barbu, J.-L. Bocquet, Resistivity recovery simulations of electron-irradiated iron: kinetic Monte Carlo versus cluster dynamics, *J. Nucl. Mater.* 352 (1) (2006) 42–49, <https://doi.org/10.1016/j.jnuclmat.2006.02.040>, proceedings of the E-MRS 2005 Spring Meeting Symposium N on Nuclear Materials (including the 10th Inert Matrix Fuel Workshop) <<http://www.sciencedirect.com/science/article/pii/S0022311506001048>>.
 - [79] E. Clouet, A. Barbu, Using cluster dynamics to model electrical resistivity measurements in precipitating AlSc alloys, *Acta Mater.* 55 (1) (2007) 391–400, <https://doi.org/10.1016/j.actamat.2006.08.021>. <<http://www.sciencedirect.com/science/article/pii/S1359645406006100>>.
 - [80] S. Golubov, R. Stoller, S. Zinkle, A. Ovcharenko, Kinetics of coarsening of helium bubbles during implantation and post-implantation annealing, *J. Nucl. Mater.* 361 (2) (2007) 149–159, <https://doi.org/10.1016/j.jnuclmat.2006.12.032>, tMS 2007: Wechsler Symposium <<http://www.sciencedirect.com/science/article/pii/S0022311506005988>>.
 - [81] J. Marian, T.L. Hoang, Modeling fast neutron irradiation damage accumulation in tungsten, *J. Nucl. Mater.* 429 (1) (2012) 293–297, <https://doi.org/10.1016/j.jnuclmat.2012.06.019>. <<http://www.sciencedirect.com/science/article/pii/S002231151200308X>>.
 - [82] T. Faney, B.D. Wirth, Spatially dependent cluster dynamics modeling of microstructure evolution in low energy helium irradiated tungsten, *Modell. Simul. Mater. Sci. Eng.* 22 (6) (2014) 065010. <<http://stacks.iop.org/0965-0393/22/i=6/a=065010>>.
 - [83] S. Krashennnikov, T. Faney, B. Wirth, On helium cluster dynamics in tungsten plasma facing components of fusion devices, *Nucl. Fusion* 54 (7) (2014) 073019. <<http://stacks.iop.org/0029-5515/54/i=7/a=073019>>.
 - [84] B.D. Wirth, X. Hu, A. Kohnert, D. Xu, Modeling defect cluster evolution in irradiated structural materials: focus on comparing to high-resolution experimental characterization studies, *J. Mater. Res.* 30 (9) (2015) 1440–1455, <https://doi.org/10.1557/jmr.2015.25>.
 - [85] M. Kirk, I. Robertson, M. Jenkins, C. English, T. Black, J. Vetrano, The collapse of defect cascades to dislocation loops, *J. Nucl. Mater.* 149 (1) (1987) 21–28, [https://doi.org/10.1016/0022-3115\(87\)90494-6](https://doi.org/10.1016/0022-3115(87)90494-6). <<http://www.sciencedirect.com/science/article/pii/0022311587904946>>.
 - [86] E. Meslin, A. Barbu, L. Boulanger, B. Radigue, P. Pareige, K. Arakawa, C. Fu, Cluster-dynamics modelling of defects in -iron under cascade damage conditions, *J. Nucl. Mater.* 382 (2–3) (2008) 190–196, <https://doi.org/10.1016/j.jnuclmat.2008.08.010>, microstructural Processes in Irradiated Materials Proceedings of the Symposium on Microstructural Processes in Irradiated Materials, as part of the annual meeting of The Minerals, Metals & Materials Society. <<http://www.sciencedirect.com/science/article/pii/S0022311508004625>>.
 - [87] A. Dunn, L. Capolungo, Simulating radiation damage accumulation in α -Fe: a spatially resolved stochastic cluster dynamics approach, *Comput. Mater. Sci.* 102 (2015) 314–326, <https://doi.org/10.1016/j.commatsci.2015.02.041>. <<http://www.sciencedirect.com/science/article/pii/S0927025615001421>>.
 - [88] A.H. Duparc, C. Moingeon, N.S. de Grande, A. Barbu, Microstructure modelling of ferritic alloys under high flux 1 MeV electron irradiations, *J. Nucl. Mater.* 302 (2–3) (2002) 143–155, [https://doi.org/10.1016/S0022-3115\(02\)00776-6](https://doi.org/10.1016/S0022-3115(02)00776-6). <<http://www.sciencedirect.com/science/article/pii/S0022311502007766>>.
 - [89] F. Christien, A. Barbu, Cluster dynamics modelling of irradiation growth of zirconium single crystals, *J. Nucl. Mater.* 393 (1) (2009) 153–161, <https://doi.org/10.1016/j.jnuclmat.2009.05.016>. <<http://www.sciencedirect.com/science/article/pii/S0022311509006424>>.
 - [90] M. Mathon, A. Barbu, F. Dunstetter, F. Maury, N. Lorenzelli, C. de Novion, Experimental study and modelling of copper precipitation under electron irradiation in dilute FeCu binary alloys, *J. Nucl. Mater.* 245 (2) (1997) 224–237, [https://doi.org/10.1016/S0022-3115\(97\)00010-X](https://doi.org/10.1016/S0022-3115(97)00010-X). <<http://www.sciencedirect.com/science/article/pii/S002231159700010X>>.
 - [91] A. Barashev, S. Golubov, D. Bacon, P. Flewitt, T. Lewis, Copper precipitation in FeCu alloys under electron and neutron irradiation, *Acta Mater.* 52 (4) (2004) 877–886, <https://doi.org/10.1016/j.actamat.2003.10.023>. <<http://www.sciencedirect.com/science/article/pii/S1359645403006396>>.
 - [92] F. Christien, A. Barbu, Modelling of copper precipitation in iron during thermal aging and irradiation, *J. Nucl. Mater.* 324 (2) (2004) 90–96, <https://doi.org/10.1016/j.jnuclmat.2003.08.035>. <<http://www.sciencedirect.com/science/article/pii/S0022311503004112>>.
 - [93] E. Clouet, A. Barbu, L. La, G. Martin, Precipitation kinetics of Al3Zr and Al3Sc in aluminum alloys modeled with cluster dynamics, *Acta Mater.* 53 (8) (2005) 2313–2325, <https://doi.org/10.1016/j.actamat.2005.01.038>. <<http://www.sciencedirect.com/science/article/pii/S1359645405000674>>.
 - [94] T. Jourdan, F. Soisson, E. Clouet, A. Barbu, Influence of cluster mobility on copper precipitation in α -Fe: a cluster dynamics modeling, *Acta Mater.* 58 (9) (2010) 3400–3405, <https://doi.org/10.1016/j.actamat.2010.02.014>. <<http://www.sciencedirect.com/science/article/pii/S135964541000100X>>.
 - [95] D. Xu, A. Certain, H.-J.L. Voigt, T. Allen, B.D. Wirth, Ballistic effects on the copper precipitation and re-dissolution kinetics in an ion irradiated and

- thermally annealed Fe–Cu alloy, *J. Chem. Phys.* 145 (10) (2016) 104704, <https://doi.org/10.1063/1.4962345>.
- [96] M. Mamivand, Y. Yang, J. Busby, D. Morgan, Integrated modeling of second phase precipitation in cold-worked 316 stainless steels under irradiation, *Acta Mater.* 130 (2017) 94–110, <https://doi.org/10.1016/j.actamat.2017.03.025>. <<http://www.sciencedirect.com/science/article/pii/S1359645417302161>>.
- [97] X. Hu, D. Xu, T.S. Byun, B.D. Wirth, Modeling of irradiation hardening of iron after low-dose and low-temperature neutron irradiation, *Modell. Simul. Mater. Sci. Eng.* 22 (6) (2014) 065002. <<http://stacks.iop.org/0965-0393/22/i=6/a=065002>>.
- [98] A. Dunn, R. Dingreville, L. Capolungo, Multi-scale simulation of radiation damage accumulation and subsequent hardening in neutron-irradiated α -Fe, *Modell. Simul. Mater. Sci. Eng.* 24 (1) (2016) 015005. <<http://stacks.iop.org/0965-0393/24/i=1/a=015005>>.
- [99] C. Varvenne, F. Bruneval, M.-C. Marinica, E. Clouet, Point defect modeling in materials: coupling ab initio and elasticity approaches, *Phys. Rev. B* 88 (2013) 134102, <https://doi.org/10.1103/PhysRevB.88.134102>. <<https://link.aps.org/doi/10.1103/PhysRevB.88.134102>>.
- [100] C. Varvenne, E. Clouet, Elastic dipoles of point defects from atomistic simulations, *Phys. Rev. B* 96 (2017) 224103, <https://doi.org/10.1103/PhysRevB.96.224103>. <<https://link.aps.org/doi/10.1103/PhysRevB.96.224103>>.
- [101] A. Dunn, L. Agudo-Merida, I. Martin-Bragado, M. McPhie, M. Cherkaoui, L. Capolungo, A novel method for computing effective diffusivity: application to helium implanted α -Fe thin films, *J. Nucl. Mater.* 448 (1) (2014) 195–205, <https://doi.org/10.1016/j.jnucmat.2014.01.039>. <<http://www.sciencedirect.com/science/article/pii/S0022311514000518>>.
- [102] A. Dunn, B. Munther, R. Dingreville, K. Hattar, L. Capolungo, Displacement rate and temperature equivalence in stochastic cluster dynamics simulations of irradiated pure α -Fe, *J. Nucl. Mater.* 480 (2016) 129–137, <https://doi.org/10.1016/j.jnucmat.2016.08.018>. <<http://www.sciencedirect.com/science/article/pii/S0022311516305967>>.
- [103] D. Xu, G. VanCoeveing, B.D. Wirth, Defect microstructural equivalence in molybdenum under different irradiation conditions at low temperatures and low doses, *Comput. Mater. Sci.* 114 (2016) 47–53, <https://doi.org/10.1016/j.commatsci.2015.11.045>. <<http://www.sciencedirect.com/science/article/pii/S0927025615007740>>.
- [104] D. Terentyev, L. Malerba, R. Chakarova, K. Nordlund, P. Olsson, M. Rieth, J. Wallenius, Displacement cascades in Fe–Cr: a molecular dynamics study, *J. Nucl. Mater.* 349 (1) (2006) 119–132, <https://doi.org/10.1016/j.jnucmat.2005.10.013>. <<http://www.sciencedirect.com/science/article/pii/S002231150500499X>>.
- [105] J.-H. Shim, H.-J. Lee, B.D. Wirth, Molecular dynamics simulation of primary irradiation defect formation in Fe10 alloy, *J. Nucl. Mater.* 351 (1) (2006) 56–64, <https://doi.org/10.1016/j.jnucmat.2006.02.021>, proceedings of the Symposium on Microstructural Processes in Irradiated Materials <<http://www.sciencedirect.com/science/article/pii/S0022311506000663>>.
- [106] D. Terentyev, L. Malerba, Diffusivity of solute atoms, matrix atoms and interstitial atoms in Fe–Cr alloys: a molecular dynamics study, *J. Nucl. Mater.* 329 (Part B) (2004) 1161–1165, <https://doi.org/10.1016/j.jnucmat.2004.04.269>, proceedings of the 11th International Conference on Fusion Reactor Materials (ICFRM-11) <<http://www.sciencedirect.com/science/article/pii/S0022311504004039>>.
- [107] D. Terentyev, P. Olsson, T. Klaver, L. Malerba, On the migration and trapping of single self-interstitial atoms in dilute and concentrated Fe–Cr alloys: atomistic study and comparison with resistivity recovery experiments, *Comput. Mater. Sci.* 43 (4) (2008) 1183–1192, <https://doi.org/10.1016/j.commatsci.2008.03.013>. <<http://www.sciencedirect.com/science/article/pii/S0927025608001560>>.
- [108] C.J. Ortiz, M.J. Caturla, Simulation of defect evolution in irradiated materials: role of intracascade clustering and correlated recombination, *Phys. Rev. B* 75 (2007) 184101, <https://doi.org/10.1103/PhysRevB.75.184101>. <<https://link.aps.org/doi/10.1103/PhysRevB.75.184101>>.
- [109] R. Stoller, S. Golubov, C. Domain, C. Becquart, Mean field rate theory and object kinetic Monte Carlo: a comparison of kinetic models, *J. Nucl. Mater.* 382 (2) (2008) 77–90, <https://doi.org/10.1016/j.jnucmat.2008.08.047>, microstructural Processes in Irradiated Materials <<http://www.sciencedirect.com/science/article/pii/S0022311508004480>>.
- [110] C. Becquart, A. Barbu, J. Bocquet, M. Caturla, C. Domain, C.-C. Fu, S. Golubov, M. Hou, L. Malerba, C. Ortiz, A. Souidi, R. Stoller, Modeling the long-term evolution of the primary damage in ferritic alloys using coarse-grained methods, *J. Nucl. Mater.* 406 (1) (2010) 39–54, <https://doi.org/10.1016/j.jnucmat.2010.05.019>, FP6 IP PERFECT Project: Prediction of Irradiation Damage Effects in Reactor Components <<http://www.sciencedirect.com/science/article/pii/S002231151000231X>>.
- [111] H. Xu, Y.N. Osetsky, R.E. Stoller, Cascade annealing simulations of bcc iron using object kinetic Monte Carlo, *J. Nucl. Mater.* 423 (1) (2012) 102–109, <https://doi.org/10.1016/j.jnucmat.2012.01.020>. <<http://www.sciencedirect.com/science/article/pii/S0022311512000244>>.
- [112] T. Jourdan, J.-P. Crocombette, Rate theory cluster dynamics simulations including spatial correlations within displacement cascades, *Phys. Rev. B* 86 (2012) 054113, <https://doi.org/10.1103/PhysRevB.86.054113>. <<https://link.aps.org/doi/10.1103/PhysRevB.86.054113>>.
- [113] A. Barbu, C.S. Becquart, J. Bocquet, J.D. Torre, C. Domain, Comparison between three complementary approaches to simulate large fluence irradiation: application to electron irradiation of thin foils, *Phil. Mag.* 85 (4–7) (2005) 541–547, <https://doi.org/10.1080/14786430412331334616>, arXiv:<http://www.tandfonline.com/doi/pdf/10.1080/14786430412331334616> <<http://www.tandfonline.com/doi/abs/10.1080/14786430412331334616>>.